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Leaving no aspect of sustainability behind: A framework for designing sustainable energy interventions applied to refugee camps

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ABSTRACT

The issues of forced displacement and energy for sustainable development are convergent: an estimated 90% of displaced people globally have no access to electricity, while 85% of refugees are hosted in developing countries. However, few tools to plan and design sustainable energy access interventions have been transposed to displacement settings. This paper presents a novel framework for the holistic planning of energy projects which considers both sustainability aspects and the specificities of displacement settings. The framework is the result of a review of literature which aimed to define a "sustainable energy intervention" in displacement contexts and an assessment of relevant planning tools against this definition. The framework includes the use of an energy delivery model toolkit, an inclusive design approach, an assessment of the desired impacts, energy system modelling, business model design, and an assessment of economic viability. We apply the framework in the design of a solar mini grid of Holl Holl refugee camp in Djibouti, in which a sustainable intervention and business model are proposed that could be compatible with the local conditions. We highlight issues that arise from the humanitarian sector status quo and propose that this framework could help to enhance sustainable energy practices in the humanitarian and development sectors.

1. Introduction

Sustainable energy access interventions are usually associated with both economic growth and social development. Affordable, sustainable, reliable and modern energy – the focus of Sustainable Development Goal 7 (SDG 7) – can act as an "engine" to directly influence productivity, income, and health, and can advance gender equality, education, and access to other infrastructure services [1]. Moreover, a recent study shows that SDG 7 cannot be achieved in isolation from other sectors due to its links with well-being, infrastructure and the environment [2]. Indeed, Fuso Nerini et al. [2] found that all SDGs and around two-thirds of the targets require action to change energy systems, including efforts to address climate change, reduce deaths from pollution, and end certain human rights abuses. Parameterising access to energy in terms of its two main components – energy for cooking, and electricity for lighting, powering appliances, and other services – highlights the scale of the challenge in achieving SDG 7 by 2030. Over 2.6 billion people rely on solid biomass, kerosene or other polluting and unsustainable fuels for cooking; the associated air pollution, mostly from cooking smoke, is linked to an estimated 2.5 million excess deaths annually [3]. Meanwhile around 770 million people globally still lack access to electricity [3], despite the socio-economic benefits it could bring.

Improving the rates of energy access will be a critical component of achieving SDG 7, and planning and designing interventions to support this will require a comprehensive understanding of the local energy needs and informed choices of technology [4]. Hiremath et al. [5] state that, as part of this planning and designing process, one must identify the sources and methods for providing energy so as to meet all of the energy requirements or demands of each task in an optimal manner. As illustrated by the term "optimal", electrification analysis and planning have most commonly relied on techno-economic methods that consider only engineering and economic criteria [6]. These approaches have been referred to as "utilitarian" by Tarekegne [4], as they exclude factors which are not easily quantifiable. Moreover, they neglect human-

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centred concerns [7], which can create poorly designed infrastructure which will have a scale or form (or both) that do not suit the social or cultural requirements of the end users [4]. It is therefore important to go beyond technical and economic measures when designing energy systems which meet the needs and priorities of its users.

The lack of consideration for local knowledge by institutions and experts has often been observed in the field of development [8–10] and this trend might be even more pronounced when it comes to displaced populations, such as refugees and internally displaced people, as suggested by Malkki [11]. The issues of displacement and energy for sustainable development are convergent: an estimated 90% of people living in displacement settings globally have no access to electricity [12], while four out of five displaced people are hosted in developing countries [13]. Displaced people living in camp situations often have levels of energy access comparable, or lower, than the nearby host communities with little or no access to electricity services for domestic or productive uses [14] and high reliance on traditional biomass for cooking, which can lead to conflicts among local populations over scarce resources [12]. Historically displacement situations have received little policy support from host governments which has limited the potential for large-scale change, while humanitarian agencies have often lacked the resources or institutional frameworks to deliver dedicated and long-term energy access projects [12,15]. Recently, however, this has begun to change.

Taking into account the overarching principle of the 2030 Agenda for Sustainable Development from the United Nations - "*Leave no one behind*" - and SDG 7, it is viewed as unacceptable by many energy access initiatives that most humanitarian organisations, with a protection mandate for refugees and internally displaced populations, do not currently provide basic energy services to displaced households [16]. With the vision to deliver SDG 7 for all displaced people by 2030, the Global Platform for Action on Sustainable Energy in Displacement Settings (GPA), hosted at the United Nations Institute for Training and Research (UNITAR) [17], was established in 2018 (formerly *Global Plan of Action for Sustainable Energy in Situations of Displacement*, renamed in 2021). This illustrates the emergence of energy as an increasingly prominent issue in the humanitarian sector; even though energy still lacks a formal administrating organisation and structure within the existing humanitarian cluster system [18] relevant actors are increasingly focused on how SDG 7 can be met in such displacement contexts.

Energy access in displacement settings is a cross-sectoral and critical issue, and the implementation of energy solutions is therefore required to deliver efficient and effective humanitarian assistance. Evidence shows that a shift from the current use of polluting diesel generators to cleaner, safer and more efficient technologies would not only reduce the costs to humanitarian organisations, but also reduce the health risks for displaced populations and reduce the environmental impacts [12]. Meanwhile, the Intergovernmental Panel on Climate Change (IPCC) recognises that climate change is likely to increase the numbers of displaced people and amplify the economic and social impacts which drive conflict and poverty over the coming century [19]. Furthermore, people living in displacement contexts are particularly vulnerable to the effects of climate change [19] and these can exacerbate tensions with the local communities who host them. The "do no harm" principle underlying humanitarian action [18], to avoid further damage or suffering as a result of its intervention, should therefore extend to the minimisation of its environmental harm and the climate-related impacts of energy provision.

Planning energy interventions in displacement settings is a complex exercise that requires careful consideration of the challenges specific to displacement settings, such as the political and legal frameworks in place locally, as policies and regulatory regimes can result in favouritism towards certain energy technologies [20]. For example, the enabling environment (including rights for refugees to work and the relationship with host communities), the supporting services (including access to finance and capacity building) and the socio-cultural context (including customary practices and local awareness) will greatly influence the way an energy system will be designed to deliver services [21]. Not only will the techno-economic aspects be affected, but also the business and ownership models of the system will be determined to some extent by these contributory external factors. The inclusion of the private sector to offer market-based solutions is advocated by development and humanitarian organisations, policy makers and academics [15], yet there is no clear guidance on how it should be integrated in the planning and design phase of an energy intervention. While tools have been developed to overcome some of the particular issues presented above in the scope of rural electrification in developing countries, few of them have been transposed to displacement settings which have their own specificities that must be considered. In this regard, it becomes imperative to plan and design energy interventions through a framework that addresses the complexities of displacement settings more systemically.

In light of these issues and arguments, this paper aims to reconcile, in the form of a framework, the two main elements that are key for planning and designing sustainable energy interventions in displacement settings. The first element consists of the holistic approach required to consider all of the key sustainability aspects of energy interventions (namely the social-cultural, environmental, political-legal, business model, economic and technological aspects described later) in the planning and design phase. The second element consists of the integration of the specificities of displacement settings that have an influence on the planning and design of energy interventions, and the supporting activities necessary for their successful implementation. This paper sheds light on how existing planning and design tools can be combined to form synergies, in order to address the sustainability challenges linked to energy systems planning and design in humanitarian context. We demonstrate how a systematic, coherent and comprehensive approach – such as the new one presented in our framework – can potentially contribute to reducing the risk of overlooking important factors, which might compromise the sustainability, implementation and adoption of the energy interventions. In Section 2 we present a literature summary of sustainability and energy interventions in displacement settings to contextualise this research and previous work in this field, and in Section 3 we describe the methods used to develop the framework and its application to a case study. In Section 4 we present a summary of the literature specific to planning and design tools for sustainable energy interventions, and outline the framework and present its development. In Section 5, we present the results of the application of the framework in theory to the case study of a solar mini grid in Holl Holl refugee camp, in Djibouti. In Section 6, we discuss the opportunities to apply this approach in other contexts and the main recommendations, while Section 7 summarises the conclusions of this paper.

2. Literature summary of sustainable energy interventions in displacement settings

2.1. Defining the sustainability of an energy intervention

The sustainability of an energy intervention can be understood in two ways. The first is by its impact on the system it exists within: the sustainability of the energy source is "a dynamic harmony between the equitable availability of energy-intensive goods and services to all people and preservation of the earth for future generations" as per the definition of Tester et al. [22,p. 5]. The second is by its longevity, with the sustainability assessing the long-term viability beyond the involvement of the project leaders, investors or donors, and its continuation despite technological changes [23]. These definitions can both be considered in terms of dimensions of sustainability. For example Bhattacharyya [24] suggests five dimensions for an energy access programme or project: *technical*, *economic*, *social/ethical*, *environmental*, and *institutional* (the latter referring to the local provision and management, ownership, staffing and monitoring). Sovacool [25] meanwhile, states that effective energy programmes consider *political*, *institutional*, *social*, and *cultural* needs in addition to *economic* and *financial* ones.

Many authors argue that several aspects of sustainability should be addressed but, in their findings, there is a greater focus on technical and quantitative factors at the expense of more qualitative socio-cultural or institutional factors. Miller et al. [26] argues that, in mid-scale projects, energy supply is almost exclusively the focus, leaving aside the socio-technical considerations which must be in place to turn this supply into energy services that deliver social benefits. Cloke [27] lays out that a top-down paradigm based on finance and technology cannot produce an understanding of which good practices will work in each circumstance and what other innovations might be required to support them. Similarly, Alfaro and Miller [28] state that ultimately economic and technical metrics should be accompanied by a more complex analytical process that accounts for *cultural*, *social*, and *economic* preferences. Cloke [27,p. 264–265] summarises the need for a holistic consideration of the range of factors: "understanding of the different *technological*, *social*, *cultural*, *economic*, *political*, *regulatory*, *research* and *organisational* arrangements… forms the basis of an energy systems literacy valuable to project implementers". These authors stress the importance of considering the multifaceted nature of sustainability, and endeavour to highlight the need to earnestly consider aspects beyond those which are straightforward to quantify.

The barriers to increased use of modern energy services among the world's poorest people, as well as the interventions that might overcome them, have also been categorised in the literature in terms of these aspects of sustainability. Watson et al. [29] reviewed 41 papers to synthesise their findings according to *economic* (both demand and supply sides), *technical*, *political* and *institutional* factors, and *cultural/social* barriers. Another piece from Akinyele et al. [30] presents a model which critically evaluates the failure factors based on the *social*, *technical*, *economic*, *environmental* and *policy* (STEEP) perspectives. Attempts to model the relationships between the *social*, *technical*, *environmental* and *economic* aspects of an energy system, such as Toba and Seck [31], suggest that the understanding of these aspects are influential and important for its planning, design, and ultimately the success of its implementation. Despite this the sustainability of the *business model*, which could capture these considerations, is rarely specifically cited as a standalone element critical for its success (or a reason for its failure) in the literature. Instead, this aspect is usually blended into the *economic*, *political*, *legal*, *policy* and/or *institutional* considerations, or divided among them. Wüstenhagen and Boehnke [32] support that innovative *business models* present a potential opportunity to overcome the main barriers for the diffusion of sustainable energy technologies. Yet, the lack of partnerships and viable *business models* prevents the scale up of *technological* innovations, especially for the bottom of the pyramid population [33]. For instance, it is often the case that even though the *technology* of an energy intervention is working, the absence of the necessary structure for maintenance, financing and continued operation – which could be synthesised in a sustainable *business model* – is an issue [34]. Successfully determining the continued value of an energy intervention, such as by using a sustainable *business model*, can increase positive feedback that enables them to provide greater value to their end users over time, rather than declining and leaving the community in poverty [26]. The *business model* itself is therefore a valuable and necessary consideration in evaluating the sustainability of an energy intervention.

The literature stresses the importance of understanding, considering and accommodating different aspects of sustainability in energy intervention planning and design. Based on this, we therefore consider a "sustainable energy intervention" as one which is:

- *socially* and *culturally* conscious of local practices, values and preferences, and including all key stakeholders in the co-design of the intervention (i.e. participatory and inclusive);
- robust, durable and *environmentally* sound (i.e. uses sustainable energy sources);
- compatible with the *political* and *legal* environment, including the formal policies and legal frameworks and existing institutional infrastructure;
- viable in terms of both the *business model* and overall *economics*, with self-sustaining revenues over time and an attractive return on investment, while solving a community need (e.g. following a social enterprise's mission);
- *technologically* adapted to the energy needs and the local environment and knowledge, while also technically feasible and optimised according to energy demand.

These six aspects (italicised) are the criteria against which we evaluate sustainability throughout this paper.

2.2. Centralised versus decentralised energy systems

Energy interventions which aim to increase electricity access, particularly in rural and remote areas of developing countries, can be broadly characterised into those which extend the existing centralised electricity network to new areas and communities and those which involve independently-operating decentralised systems [35]. The former, implemented via centralised electricity network planning, has typically been followed by countries trying to rapidly enhance electricity access but could be a contributor to the present deficit in electricity access, as it necessitates significant upfront infrastructure investments in order to take advantage of economies of scale at utility-scale coal, natural gas, nuclear, or hydroelectric power plants [36]. However, planners of centralised electricity systems and the extension of national grid networks often overlook or do not accommodate the needs of rural and remote areas and poor communities, for example owing to comparatively low demand by the users or relatively high costs to the installers, and has led to negative environmental impacts from fossil fuel consumption and forest degradation [5]. The centralisation and the reliance on fossil fuels can result in high costs of electricity and greenhouse gas (GHGs) emissions, which raises concerns about climate change and health hazards. It has therefore become critical to find alternative systems of power generation and distribution [37].

Decentralised electricity systems, and the growing number in recent years that are mostly based on available renewable energy sources, offer a solution for remote locations by generating power closer to the demand site. While they usually operate at smaller scales than central utility plants, these systems can supply electricity in a more reliable and environmentally friendly way to remote communities [37]. A decentralised approach often makes the systems which rely on local renewable sources a more suitable intervention for rural electrification, compared to fossil fuel and grid extension alternatives, especially in rural and remote areas of developing countries where extending the national grid network would be prohibitively expensive. Furthermore, a connection to the national grid may be technically possible but not legally permitted owing to government policy or regulatory restrictions. In some countries this, explicitly or implicitly, precludes residents of refugee camps and other displacement settings from connecting to the national grid as the supposedly temporary nature of their situation would not allow them to access to a solution that is perceived as permanent [38] Depending on the specific nature of each displacement setting, decentralised systems may therefore be the most cost-effective, and in some cases only, option for electrification.

2.3. The state of energy access in displacement settings

The issue of energy access in emergency situations has been gradually included in the global humanitarian response since the beginning of the 21st century [21]. The recognition of the importance of the environmental protection agenda by humanitarian agencies has allowed the energy topic to be considered. This has been supported by energy issues also being addressed through the focus on beneficiaries' well-being and protection concerns [21]. The first publication raising awareness on energy issues related to camp set-up and the harmful overuse of firewood for cooking dates from 1995 [39] and the recommendations highlighted in the article — including early planning for energy needs, fuel availability assessments, energy conservation (lids, wood drying and heat-storage cookers), stoves and reforestation — remain relevant to this day.

At the end of 2020, there was an estimated 82.4 million forcibly displaced people worldwide as a result of persecution, conflict, violence, human rights violations and events seriously disturbing public order [40]. Refugees represented 26.4 million people [40] and 55 million were internally displaced people across the world, 48 million as a result of conflict and violence, and 7 million as a result of disasters [41]. Today, more than 60% of all refugees and 80% of all internally displaced populations (IDPs) are living in urban areas, while 30% of all refugees live in rural camps [42]. Developing countries host the majority of the forcibly displaced populations (four out of five people [13]) who usually have similar, or worse, access to energy as their host communities. Moreover, displaced populations caught in protracted crises are often excluded from national energy development strategies and activities, despite the fact that the potential benefits of including them are highlighted in the nascent literature on humanitarian energy [15].

The amount of data on the level of energy access in displacement settings is extremely limited [18]. However, global estimates indicate that 80% of the refugees and displaced people in camps have absolutely minimal access to energy, with high dependence on traditional biomass for cooking and no access to electricity [12]. Moreover, it is estimated that 89% of displaced populations hosted in refugee camps have access to only the most basic forms of lighting (such as candles, kerosene lamps, or nothing at all [12]) and as many as 7 million displaced people in camps have access to electricity for less than 4 h [43]. The utilisation of natural resources, such as firewood, agricultural land, water sources, and animals, by displaced populations can impact both the relationships they have with the host communities as well as the environment. Examples of tensions arising from deforestation and competition over scarce resources can affect the relations of refugees and their hosts: some recent concerns about the increasingly violent nature of refugee-host coexistence were highlighted in Uganda [44], and more specifically explored by Miller and Ulfstjerne in the case study of the Rhino Camp [16]. Finally, the increase in demand ensuing the creation of settlements for displaced population can not only create negative social and environmental impacts, but also additional waste that can also be a threat to human health [45].

Several international initiatives aim to address energy needs in displacement settings, both by providing local energy solutions as well as setting a common operational framework for the sector to transition to cleaner energy solutions [38]. Some examples of global sectoral initiatives are the Moving Energy Initiative [46], the UNITAR-led Global Platform for Action on Sustainable Energy in Displacement Settings (GPA) [17], or the UNHCR Global Strategy for Sustainable Energy [47] and its Clean Energy Challenge [48]. The Renewable Energy for Refugees (RE4R) Project from Practical Action, funded by the IKEA Foundation, and implemented in Rwanda and Jordan between 2017 and 2022 is one example of local initiatives to improve the health, wellbeing and security in camps and host communities through access to sustainable energy. Another large-scale project called Enter Energy was announced by Shell in 2019 to support energy access for refugees and displaced people and their host communities in Mozambique and Ethiopia, among other countries [49]. These recent examples highlight the emerging prominence of the need to address energy issues in displacement settings, but thus far the levels of support afforded to it lag behind the large and growing needs.

2.4. The specificities of planning and designing energy interventions in displacement settings

Even though providing energy in displacement settings is complex and challenging, it is a key facilitator for delivering primary humanitarian responses in an efficient manner [18]. For instance, energy is required to pump and purify water for drinking and cooking, which is linked to the water, sanitation and hygiene (WASH) response area [50]; energy is also vital for food security through preparation and preservation, and as a means to generate income as a livelihood strategy [50]. Moreover, sustainable energy sources can mitigate the need for fuelwood collection in remote and unsafe locations, especially by women and girls [51]. Lighting is also beneficial for livelihood activities during dark hours as well as for protection from sexual and gender-based violence (SGBV) and animal attacks [52]. Finally, electricity and fuel are required for logistical support, from the transport and delivery of basic goods to the provision of healthcare and communications [18].

Despite energy being a fundamental enabler for these core humanitarian functions, many institutional and organisational challenges stifle its integration into operational planning. The short-term annual budgetary cycle of humanitarian response prevents long-term investment in energy infrastructure, whose payback periods are usually several years, rather than just the one offered by typical humanitarian planning horizons [15]. Displacement settings are also perceived as risky for investment and operation by the private sector, which could offer more flexibility for long-term financing to help overcome short-term budget cycles [53], and this is exacerbated by a lack of enabling policies or regulatory clarity which inhibit its engagement in delivering energy services [53].

Another barrier for the inclusion of energy in humanitarian programming lies in the fact that energy is not represented as a specific cluster within the humanitarian response [54]. This results in shared but often uncoordinated responsibility for energy supply infrastructure and fuel among humanitarian partners and ultimately a lack of accountability [18]. Projects which focus on just one area of energy access may leave other needs being unmet, unless specific additional attention is given to them: providing electricity access in households would likely not address issues relating to cooking unless, for example, electric cookers were also successfully supported. Additional consideration of potential future uses of energy, for example electricity for electric vehicle charging for the use of humanitarian organisations, are also unlikely to be met under projects which focus on specific or short-term goals.

The humanitarian sector is currently relying on electricity practices to power their infrastructure that are costly, environmentally harmful and ineffective in meeting the needs of displaced people [17]. A recent survey conducted by the GPA Coordination Unit highlights that, based on feedback from six United Nations organisations and the International Committee of the Red Cross (ICRC), there are currently more than 11,000 generators in use in humanitarian settings around the world [55]. The initial conservative estimates are that humanitarian agencies spend more than USD 100 million on petroleum fuel per year, emitting approximately 200,000 t of CO_2 [55]. In addition, it was evaluated that a total investment of around USD 250 million would be necessary to solarise viable systems, which could save around USD 70 million in fuel costs and around 125,000 t of CO_2 per year [55]. The use of diesel generators is mainly mainstreamed for humanitarian agencies in off-grid areas and places where the grid is unreliable, whereas electricity products and services for displaced communities themselves are often provided for free through distribution. The latter reinforces the dependency of displaced populations on continuous aid [52] and can disrupt existing livelihoods and markets [56]. Recently, a nascent shift from an aid-based to a market-based approach has been observed in regard to energy access through private sector delivery, benefiting both refugees in camps and their host community [52]. A similar trend has been noted regarding inclusive programming in which displaced people and host communities are actively involved in the development, design, implementation and evaluation of the energy intervention [54], as opposed to a top-down approach.

These emerging trends suggest that there is an indisputable need to address the issue and find better solutions that reconcile the *socio-cultural*, *environmental*, *political*, *legal*, *business model*, *economic*, and *technological* aspects of sustainable energy interventions in these contexts. Energy needs can be assessed and subsequently addressed through specific *technological* energy solutions, and their expected technical performance, yet it is challenging to navigate the complexities and dimensions of the planning and design stages of projects and programmes in displacement settings. Recent case studies on energy in displacement settings (mostly refugee camps) [38,57–60] have partially addressed some of these issues, but more work is necessary to explore these issues fully. To date, no comprehensive framework has been suggested to provide such guidance. We therefore present a methodological approach to consider the six sustainability aspects, which aims to integrate the specific considerations that are particular to humanitarian aid and decrease the risk of failures of the interventions. We suggest a holistic framework that recommends an inclusive approach, a definition of the cross-sectoral benefits, a *techno-economic* and *environmental* assessment of the energy system, a sustainable *business model* design, and an evaluation of the long-term viability of the intervention from the private sector's perspective.

3. Methods

3.1. Methods to develop the framework

This paper presents a framework developed by reviewing the literature related to the planning and design of energy interventions (Section 4.1), identifying the limitations of different tools and methods presented in it, and highlighting their respective complementarities and synergies. This literature review encompassed the tools to implement inclusive and participatory design approaches, to assess the potential impacts and transformative changes within communities, to design sustainable business models, to evaluate the economic viability and to design optimal renewable or hybrid off-grid or on-grid electricity systems. We considered tools presented in academic and practitioner literature that were not limited to only those used in displacement settings or developing countries, or for planning and designing energy interventions, but more general project design, management, and modelling tools. No relevant materials were explicitly excluded. Particular attention, however, was given to those that have been used and recommended for the Global South and/or refugee camps.

From this review of the literature, the key questions that the planner or designer should address were identified in the light of the six sustainability aspects. In addition, we evaluated the aims and outcomes of each tool to identify its contributions to the planning and design of sustainable energy access intervention, and then the step in the overall process in which each it is used. Then, we assessed the limitations in scope of each tool as applied to the six aspects. Finally, we identified the complementarities and synergies from the combination of tools in order to propose a holistic framework.

3.2. Methods for applying the framework to the case study

In this subsection we outline the methods used to apply the framework, presented in Section 4.2, to the case study of the solar mini grid in Holl Holl refugee camp in Djibouti, the results of which are presented in Section 5.

In a first instance, a literature review specific to sustainable energy and energy interventions in Djibouti was conducted in order to contextualise the research and acquire an understanding of the relevant topics, as well as to be aware of previous studies in the field. Then, the application of the framework consisted of a mixed-methods approach using a combination of qualitative and quantitative analyses. Due to the COVID-19 global pandemic, it was not possible to conduct a field trip to Holl Holl refugee camp in Djibouti to collect primary data in the form of household surveys and key informant interviews, as initially planned. While it was not feasible, it is key to stress that ideally this kind of direct engagement under more regular favourable circumstances is of the highest importance to favour an inclusive approach, which is unfortunately frequently overlooked or done only superficially. However, the results of an assessment conducted by one of the main project partners, SELCO Foundation, were shared. It provided an overview of the needs in the three refugee camps of Djibouti, including Holl Holl, based on short field visits allowing for observations and some interviews of key informants. As a complement, semi-structured interviews of different local stakeholders - mostly humanitarian organisations - were performed, recorded and analysed using a rapid thematic/domain qualitative analysis to inform the research and collect secondary qualitative data (see Section A in Supplementary materials for more details). The literature review and desk research combined with the interviews of the key stakeholders enabled us to define the mandates, missions and priorities of the different organisations and community groups in the scope of the intervention.

To complete the data collection, an extensive review of the existing reports, both public and internal, from humanitarian organisations, development actors and international organisations active in displacement settings and rural areas in Djibouti was undertaken. Then, secondary quantitative data were obtained from a number of different open-source databases and previous mini grid studies. These data allowed us to estimate the energy demand of the institutional and business uses in the refugee camp. Some were also used as inputs for the technical, financial and environmental modelling of the solar mini grid. The lack of primary data on the electricity demand in the case of Holl Holl refugee camp resulted in the need to estimate the load profile necessary as an input to the energy modelling software. It was chosen to follow the method suggested by Sacino et al. [61], Sacino [62] and Ziade [63] (see Section B in Supplementary information for further details). In addition, the pros and cons of each type of mini grid ownership models were weighed by using guidance from the literature [64–68]. Finally, recommendations and best practices from the literature specific to displacement settings [69–71] were used to inform the choice of ownership model.

4. Development and presentation of the framework

4.1. Review of the tools for the planning and design of a sustainable energy intervention

Project planning, in its essence, is a way to reduce risk exposure [72]. Moreover, the sustainability of an energy intervention is in the first instance influenced by the degree to which the *social-cultural*, *environmental*, *political-legal*, *business model*, *economic* and *technical* aspects are considered during its planning and design phases. Incorporating these sustainability considerations during these phases should therefore allow for better risk management and reduce the likelihood of failure.

Planning and designing a sustainable energy intervention consist of several key steps. One of the preliminary and central activities is to conduct an energy needs assessment in order to collect data about current energy access, expenditures, and aspirational energy needs [73]. Existing toolkits have been developed, such as those from D-Lab at the Massachusetts Institute of Technology [74] or SELCO Foundation [75]. These assessments provide fundamental inputs for the energy system *and business model* design, as well as for the identification of the supporting activities required to create an enabling environment for the success of the intervention. They also are a way to capture the opinions and suggestions of the local authorities and the community targeted by the intervention to promote a bottom-up approach and lay the foundations for a participatory and inclusive project design [73]. Participatory and inclusive approaches are based on the assumption that common decision making and engaging with stakeholders allows for a sense of ownership and empowerment, which enables more sustainable social impact [7]. Several methods, tools and frameworks can be found in the literature regarding rural energy planning practices for development and for humanitarian planning. A common approach consists in mapping the influence and relationships of the stakeholders as well as their interests and importance [76,77].

As a complement to the energy needs assessment, the framework for Sustainable Energy Access Planning (SEAP) developed by the Asian Development Bank suggests a benefit assessment to evaluate the positive impacts associated with energy access at the micro and macro levels [78]. A common model used in the non-profit sector to map out such benefits is the programme theory (also called *Theory of Change* or ToC) which describes the inputs, activities, outcomes and impacts of an intervention in contribution to meeting its overall goals [79]. The development of visual depictions was led by United Way and the W.K. Kellogg Foundation in the late 1990s and resulted in the *Logic Model Canvas*, also referred as the *Impact Canvas* [80]. Today, the *Impact Canvas* is used as part of the methodology of some research centres focused on sustainable solutions for humanitarian contexts, such as the EPFL EssentialTech Centre [81].

Another key step of the planning and design phase is the modelling of the energy system. Based on the choice of *technology*, the energy (or specifically electricity, as is more common when using numerical modelling for community-scale systems) demand and the desired impacts, the system can be designed using software to computationally model its expected performance. Several tools are available to simulate and optimise both on and off-grid power systems. *HOMER* is one such software and is among the most popular commercial tools for mini grid system design. An alternative open-source software, called *CLOVER*, was developed by the Grantham Institute at Imperial College London to simulate and optimise mini grid systems [82–85] and provides flexibility to the user via its editable code to meet specific design requirements. It can account for potential future growth in the electricity demand or expansion of the system capacity, as well as the GHG emissions linked to the embodied energy of the technologies. These two features are of key interest when modelling sustainable energy interventions – for example solar mini grids – as they capture the need for long-term viability and the *environmental* aspect.

Once the *technical*, *environmental* and *economic* characteristics of the optimal energy systems are defined, the next step is to design a *business model* that is aligned with the sustainability objectives of the intervention. Generic *business models* are usually built with the help of the *Business Model Canvas* which is a visual chart with the four pillars that describe a company's product or service, customers, infrastructure and finances [86]. As an alternative, the *Social Business Model Canvas* (SBMC) is an adaptation of the *Business Model Canvas* from Osterwalder et al. [86] and has been revised to reflect the sustainability factors of a venture or project that aims to solve a social issue in low or middle-income countries [87]. In summary, the *SBMC* is divided into three parts. The top segment enables the social enterprise to define their social business strategy, mission and values, as well as their intended impact and identification of the beneficiaries. The middle segment is concerned with the business operations and revenue generation aspects. While the bottom part provides an overview of the performance of the organisation with respect to *economic*, *environmental* and *social* aspects, sometimes referred to as the Triple Bottom Line [88].

Finally, ensuring the *economic* viability of the energy intervention in the light of the energy system design and *business model* is an essential step. A simple and common approach is to carry out a *Profit and Loss (P&L)* analysis, where capital expenditures (CAPEX) and operational expenditures (OPEX) are forecasted and compared to the expected savings from the project or programme, or most likely the revenues from the business.

4.2. Designing the holistic framework

4.2.1. Identification of the steps, key questions, and tools for the framework

The first outcome of this literature review was the identification of five distinct steps necessary for planning a sustainable energy intervention: “Inclusive design approach”; “Assessment of the desired impacts”; “Energy system modelling”; “Business model design”; and “Economic viability assessment”. As part of this mapping, we also identified and summarised the key questions from the literature which are addressed in each step of the planning and design process in the light of the six sustainability aspects, as presented in Fig. 1. The steps are followed sequentially from top to bottom as part of the planning process, but returning to previous steps and revisiting questions as part of an iterative process may be required as more information becomes available throughout the planning stages.

The next step in our aim to develop a holistic and systematic framework was to determine the limitations in scope of each tool as applied to the six aspects (*social-cultural*, *environmental*, *political-legal*, *business model*, *economic*, and *technological*). As shown in Table 1, this identified the inability of any single tool to sufficiently capture all the aspects and enables us to see how several might be used together in a complementary way.

As shown in Table 1, several tools have been identified to support the planning and design of sustainable energy access interventions. Some tools, such as the *Stakeholder (Influence) Mapping Tool* [76,77], the *Logic Model/Impact Canvas* [80,81], the *Theory of Change* [96], the *Business Model Canvas* [86,99], and the *Profit and Loss Forecast* [102,103], are well-established and generic tools used in project management and business development. Some other sources have been specifically revised to incorporate a sustainability aspect, such as the *Triple Layered Business Model Canvas* [88] and the *Social Business Model Canvas* [87,100]. Additional tools focused on inclusive humanitarian planning (but not energy access) are for example *Bottom-Up People-Centred Approach to Humanitarian Planning* [89,90] and the *NRC Community Coordination Toolbox* [95]. Finally, specialised tools have been recently designed to reconcile sustainable and inclusive energy access, such as the *Embedded and Inclusive Programming* [91] and the *Inclusive Energy Access Handbook* [93]. Even though this list is non-exhaustive, it provides the necessary foundation for the framework to be developed.

4.2.2. Identifying complementarities and synergies from the combination of tools

By considering the step of the planning and design process that the different tools shown in Table 1 contribute to, and their potential complementarities, it is therefore possible to determine how using several of them can create synergies and inform sustainable energy intervention design in greater depth. We found the most complete tool, in terms of the number of aspects covered, was the *Energy Delivery Model (EDM) Toolkit* approach by Garside and Wykes [92] and so used it as the foundation of our wider framework. As the *EDM* focuses on poor groups living in developing countries it is relevant to include additional consideration regarding inclusive design specific to displacement settings: we therefore included the *Embedded and Inclusive Programming* framework for humanitarian energy services, from Rosenberg-Jansen et al. [91], as an enhancement to the *EDM* approach.

A limitation of the *EDM Toolkit* is that the *Problem Tree* and *Problem Solution* tools suggested to identify the fundamental causes of the problem and their most important effects do not specifically address the *social*, *economic* and *environmental* aspects. As an alternative, we included the *Impact Canvas* (and the underlying *Theory of Change* mapping associated with it) as an appropriate tool to deepen the understanding of these aspects. Similarly we selected the *Social Business Model Canvas* as a visual tool to capture these aspects, an improvement on the generic *business model* canvas from Osterwalder et al. [99] suggested in the *EDM* approach.

A further limitation of the *EDM Toolkit* is the lack of guidance on the

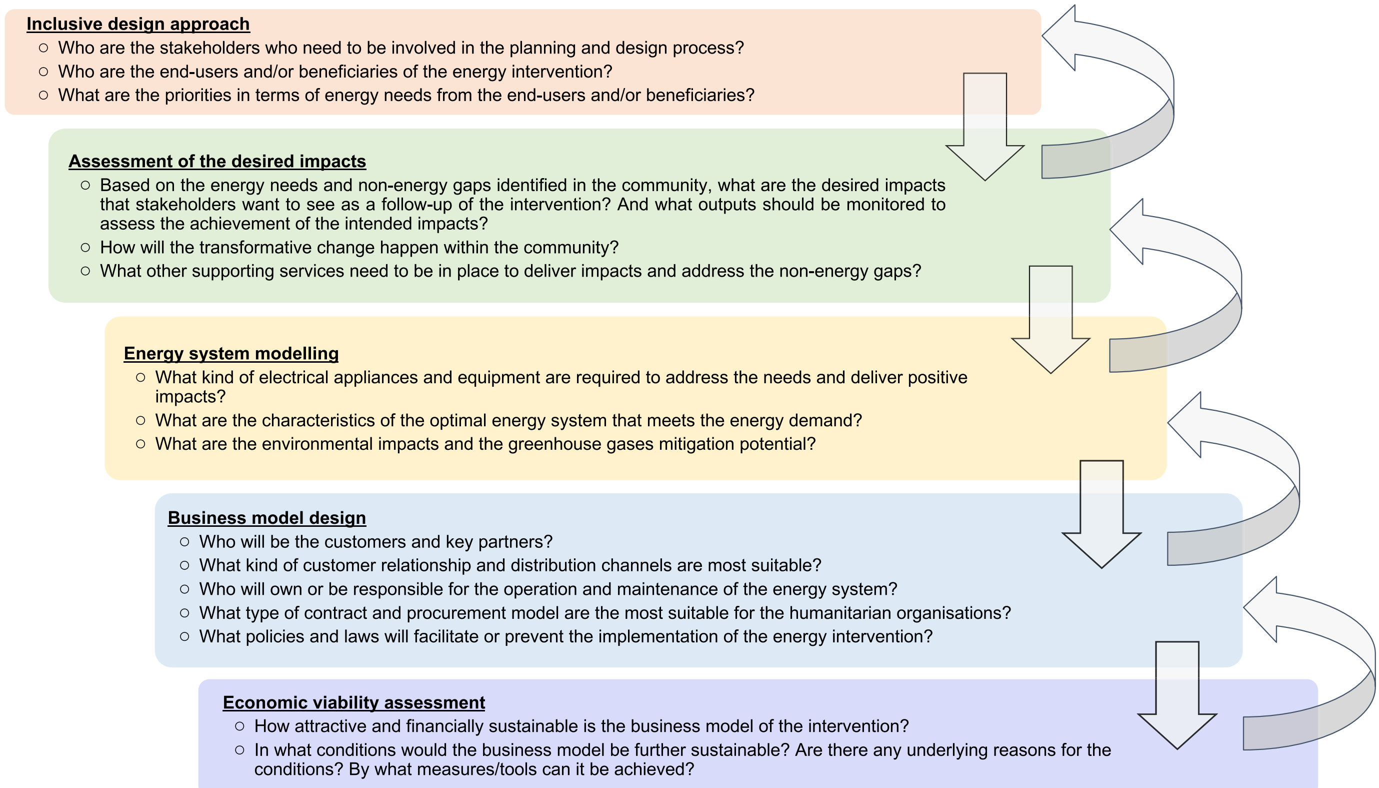

Fig. 1. Key questions for the planning and design phases of a sustainable energy access intervention.

Table 1
List of tools reviewed, and their associated aspects considered (✓ = aspect considered; x = aspect not considered; (x) = limited consideration of the aspect).

Step of the planning and design	Name of the tools	Aspect(s) considered						References
		Social-Cultural	*Environmental*	*Political-legal*	*Business model*	*Economic*	*Technological*	
Inclusive design approach	*Bottom-Up People-Centred Approach to Humanitarian Planning*	✓	x	x	x	x	x	[89,90]
	Embedded and Inclusive Programming	✓	x	(x)	(x)	(x)	(x)	[91]
	Stakeholder (Influence) Mapping Tool	✓	x	✓	(x)	(x)	(x)	[76,77]
	Energy Delivery Model Toolkit	✓	✓	✓	✓	✓	(x)	[92]
	Inclusive Energy Access Handbook	✓	(x)	(x)	x	(x)	(x)	[93]
	Problem Assessment Toolkit	✓	x	x	✓	✓	✓	[94]
	NRC Community Coordination Toolbox	✓	x	(x)	x	(x)	x	[95]
Assessment of the desired impacts	*Logic Model/Impact Canvas*	✓	(x)	(x)	x	(x)	x	[80,81]
	Theory of Change	✓	(x)	(x)	x	(x)	x	[96]
Energy system modelling	*HOMER Software*	x	✓	x	x	✓	✓	[97]
	HOMER Powering Health Tool	x	✓	x	x	✓	✓	[98]
	CLOVER Software	x	✓	x	x	✓	✓	[83,85]
Business model design	*Business Model Canvas*	x	x	(x)	✓	✓	x	[86,99]
	Triple Layered Business Model Canvas	✓	✓	(x)	✓	✓	x	[88]
	Social Business Model Canvas	✓	✓	(x)	✓	✓	x	[87,100]
	Lean Canvas	x	x	(x)	✓	✓	x	[101]
Economic viability assessment	*Profit and Loss Forecast*	x	x	x	✓	✓	x	[102,103]

technical design of the proposed *technological* energy solution: there is no approach recommended for the modelling of the energy system, which can have a great influence downstream for the *business model* design and the *economic* viability assessment. In our case for mini grid simulation and optimisation we use the *CLOVER* software as a suitable tool owing to its flexibility and ability to compute the costs and GHG emissions of electricity systems, and to explore options for mitigation. Moreover, its functionality to account for future growth in community electricity demand is much aligned with the goals of sustainable development, such as in the ESMAP Multi-Tier Framework [14]. With improvements in the energy efficiency of appliances, however, increases in access to electricity services may be decoupled from increases in power consumption. This is also aligned with the *CLOVER* model, which parameterises usage first in terms of demand for electricity services, and subsequently the resultant electricity demand, and allows for comparison between appliances with different power ratings in providing the same or similar services.

Finally, the assessment of the *economic* viability of the energy intervention is not thoroughly addressed in the *EDM Toolkit* since no specific tool is recommended to support the evaluation, apart from a simplified accounting exercise at the “top line” level [92]. As this is a key part for a sustainable *business model*, we identify a *Profit and Loss Forecast* as an essential complement in the planning and design phases.

4.2.3. The resulting framework

The identification of the steps and key questions of the process, and the complementarities and synergies of the tools allowed us to develop a systematic approach. Fig. 2 illustrates the holistic framework which maps our combination of tools that cover all six aspects that influence the successful planning and design of a sustainable energy intervention. The figure highlights how each tool (rectangles) is used to inform other components of the framework (arrows pointing in the direction of information flow), with the aspects (squares) they inform at the bottom.

The *EDM Toolkit* shown at the top of Fig. 2 is the most comprehensive tool used in the framework since it covers most of the sustainability aspects. It is used as a foundation and therefore is used in parallel of the other tools. Thus, it does not belong to a single step but the outcome of using the tool rather informs and influences the rest of the steps. The use of the *Embedded and Inclusive Programming* approach allows the planner to identify the necessary information necessary to collect from the stakeholders, including their needs, and about the environment and markets. This then informs the "Assessment of the desired impacts", as well as the "Business model design". Once the long-term objectives are formulated thanks to the *Impact Canvas*, the "Energy system modelling" can be undertaken as the energy needs, and how they should be met, are clear. Then, the results of the optimal sizing are leveraged (in combination with the outcomes of the tools previously described) to inform the *environmental* impacts of the "Business model design" including the choice of ownership and procurement models. Finally, the financial impacts of the technical model also flow to evaluate the "Economic viability assessment" of the energy system. Overall, the use of the tools within this framework allows the planner to design of the energy intervention by incorporating all of the six sustainability aspects and keeping a holistic view during the whole process.

As much information gathered is relevant to different tools, then it is likely that several tools should be used concurrently, rather than consecutively, and be developed iteratively as the available evidence supporting the intervention grows. Therefore, even though iterations are not shown explicitly in Fig. 2, the user of the framework should anticipate the need for revisiting the previous steps based on the new information available over time. Fig. 1, however, highlights the potential need for an iterative process in addressing the key questions. For instance, involving multiple stakeholders in a complex and continually evolving context throughout the project's inclusive planning and design may require the user to set aside time to adjust the intervention based on the feedback received.

5. The illustrative case study of Holl Holl refugee camp

5.1. Background information on the case of Holl Holl refugee camp

5.1.1. Energy and humanitarian affairs in Djibouti

The Republic of Djibouti is a small country located in the Horn of Africa. It shares its borders with Somalia in the south, Ethiopia in the southwest, Eritrea in the north, and the Red Sea and the Gulf of Aden in the east, with Yemen across it. Its population is approximately 1 million inhabitants of which 78% live in urban settings [104]. Djibouti currently hosts over 31,300 refugees and asylum-seekers – about 3% of the country's population – mostly from Ethiopia and Somalia [105] and has a tradition of hosting refugees spanning over four decades. The Government of Djibouti has a relatively open policy towards displaced people and adopted the Comprehensive Refugee Response Framework (CRRF) in 2016, which has the goal to strengthen the nexus between humanitarian and development actors by encouraging joint responses [106]. The CRRF aims to build self-reliance and resilience, and to lay the basis for sustainable solutions, which creates a conducive environment for energy transition projects [107].

The electricity infrastructure in Djibouti is unreliable: voltage fluctuations, electrical spikes, blackouts, brownouts and other disruptions have significant impacts on the operational effectiveness of

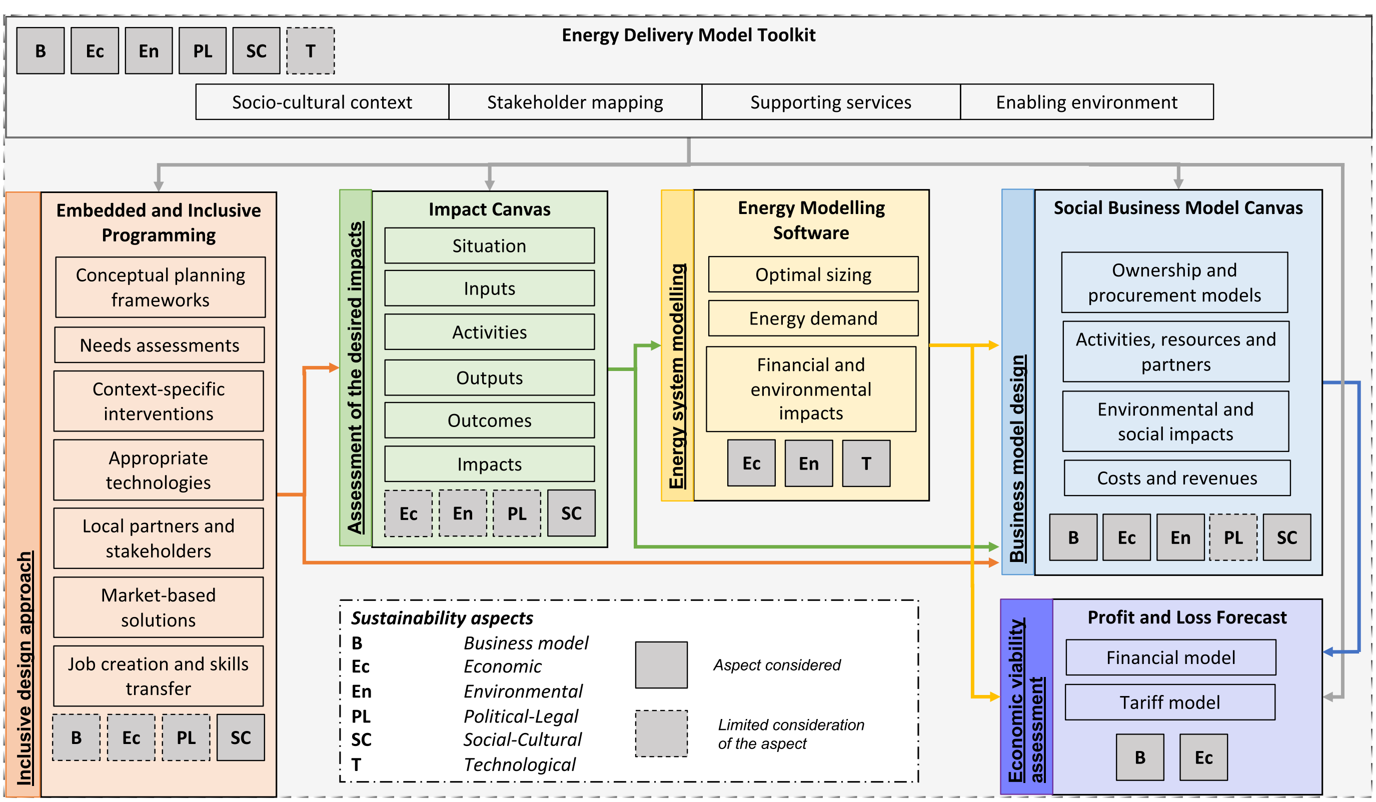

Fig. 2. A holistic framework for planning and designing sustainable energy interventions in displacement settings.

humanitarian activities as well as the household level electricity access [108]. Moreover, high electricity prices and diesel costs induce high electricity bills. For organisations like UNHCR, the UN Refugee Agency, this translated into spending approximately USD 3 million on electricity and diesel at its offices and camps in 2017 [108]. In 2014, the Djiboutian government launched a long-term development plan, Vision 2035, with one aim to transition from "100% thermal" energy based on combustion of fossil fuels to "100% renewable" sources by 2020 [109]. However, the latest data available by 2020 from the International Renewable Energy Agency (IRENA) show that despite a 21% increase in installed renewable electricity capacity between 2015 and 2020, the electricity generated in 2019 was >99% from non-renewable sources [110]. This target was expected to be possible to meet via abundant and underexploited solar, wind, and geothermal resources [109], which together could generate more than 300 MW of electric power [111]. Meanwhile off-grid renewable systems have the potential to meet the electricity demand in unserved rural areas in Djibouti, which are far from the existing grid network and can replace existing diesel systems [107]. Implementing efficient, off-grid renewable energy solutions could therefore potentially offset costs to humanitarian agencies as well as the rural and displaced populations while aligning with the national policies of the country.

Despite this technological potential, the legal framework for the implementation of an independent power producer (IPP) law is slow, even though it is supported by the Power Africa partnership led by USAID [111]. The national electricity company, Electricité de Djibouti (EdD), has the monopoly on supply and, at present, a concession from the Government is required to sell electricity. Furthermore, tariffs must be aligned with the grid levels unless subsidies are arranged between EdD, the national development agency Agence Djiboutienne de Développement Social (ADDS), and/or the Ministry of Energy and Natural Resources. Finally, as in many other displacement contexts, humanitarian organisations responsible for refugee camp coordination and management (e.g. UNHCR) cannot sell electricity directly to the refugees. This means they have to outsource this service to a third party, such as the private sector, and are usually unwilling to own long-term electricity systems if they do not serve their own facilities and operations.

5.1.2. Holl Holl refugee camp

Holl Holl refugee camp is located in the South-East region of Djibouti, 53 km from the capital Djibouti Ville, and is not connected to the national grid. Holl Holl is accessed by an unpaved road which makes the supply of fuel and delivery of goods, services and humanitarian assistance challenging. The camp was closed in 2006 but reopened in 2012 to decongest the refugee camp of Ali Addeh, and is home to 6359 refugees from Somalia (51%), Ethiopia (41%) and Eritrea (7%) [112]. It is 2 km away from Holl Holl Village, which has a population of approximately 3000 Djiboutians, and has had access to the grid since 2019. Holl Holl became an administrative and commercial centre in the 1900s after the construction of the railways between Addis Ababa (Ethiopia) and Djibouti Ville but the activities in the area have since declined, inducing high levels of unemployment.

Since July 2019, The Global Platform for Action Coordination Unit (GPA CU), hosted by UNITAR, has embarked in a project that aims to support United Nations (UN) humanitarian agencies active in Djibouti (such as UNHCR, WFP, FAO, and IOM) transition to sustainable energy solutions [108]. In addition, it aims to leverage the provision of energy access to displaced populations to support income generating activities, among other objectives. The project identified the energy needs in Holl Holl camp and scoped an initial assessment of the goals and specifications of a potential sustainable electricity intervention, suggesting the use of a solar mini grid to supply power to the primary school, the clinic, seven offices, 40 streetlights, and potentially the local marketplace. The cost was estimated at approximately USD 80,000 (excluding any household connections) and compared to the cost of grid extension from the neighbouring village, suggested by ADDS, at an estimated cost of over USD 180,000 (not including internal wiring, streetlights and electricity bills) [107]. Even though the electricity access at the household level was identified as a gap, the poor quality of the roofs of the shelters is a barrier for the installation of any heavy electricity systems (e.g. solar panels) or low voltage connection lines with sufficient safety measures. For that reason, connecting households to the solar mini grid was abandoned and not consider further in the planning and design phase.

Concurrently SELCO Foundation, another lead partner of the project, undertook an Ecosystem Assessment [113,114] of the displacement settings in November 2019 to identify gaps and develop solutions to support refugee livelihoods and energy access. This high-level assessment gave an overview of the needs in the three refugee camps of Djibouti, including Holl Holl, based on short field visits allowing for observations and a few interviews of key informants. Their findings included suggestions for entrepreneur training, financial access programmes and business improvements which could be made possible via access to electricity [113,114].

The work of the GPA CU and SELCO Foundation made important steps towards scoping an energy intervention that would meet pressing needs in Holl Holl camps but was limited by time, resources and external factors to develop the intervention any further. Aside from the overarching objective of providing electricity to the community and humanitarian facilities to reduce the dependency and expenditure on diesel, as well as the businesses to improve income-generating activities, no predefined outcomes and impacts were clearly articulated for this proposed intervention. Some activities were drafted however, such as the deployment of some technology, building financial systems, and training and incubation of energy enterprises. Moreover, no energy needs assessment at the community, institutional and business levels were conducted prior to the start of the study, and earlier assessments by UN agencies were mostly outdated. In addition, the ownership and procurement models for the solar mini grid were only briefly factored out in the light of the political and legal context regarding Djiboutian electricity systems. Finally, the recommended choice of business and financial models for operating the system and providing power to the business users was not crafted.

The case of Holl Holl camp therefore offers an opportunity to demonstrate the viability of the framework we present in this paper. The local context, energy needs, and potential objectives have been scoped for Holl Holl by the GPA CU and SELCO Foundation, albeit at a relatively high level, but many factors remain unknown and there would be insufficient information to proceed with the deployment of an energy intervention. Furthermore, the sustainability of the potential mini grid would be highly uncertain given the status of the proposal after the initial project. We therefore build on their analysis and apply our framework to a theoretical application for Holl Holl refugee camp, using it as a case study to test its ability to address six core aspects which influence the success of an energy intervention.

5.2. Application of the framework to the case study

5.2.1. Inclusive design approach

As highlighted in the *EDM Toolkit* and the *Embedded and Inclusive Programming* approach, the engagement of the stakeholders in the design of the energy services is crucial in the pursuit of a participatory approach [91], as well as for understanding the *socio-cultural* context and local dynamics of the situation in which the intervention will be implemented. The first step of the framework enabled us to understand who the stakeholders of the solar mini grid project in Holl Holl refugee camp are, their interests and level of influence, and how to engage with them.

An overview of the stakeholders relevant to the Holl Holl project is presented in Fig. 3, following the models presented in the stakeholders mapping and analysis tools [76,77]. Desk research combined with the consultation of the key stakeholders enabled us to better understand the mandates, missions and priorities of the different organisations and community groups in the scope of the intervention. Furthermore, an

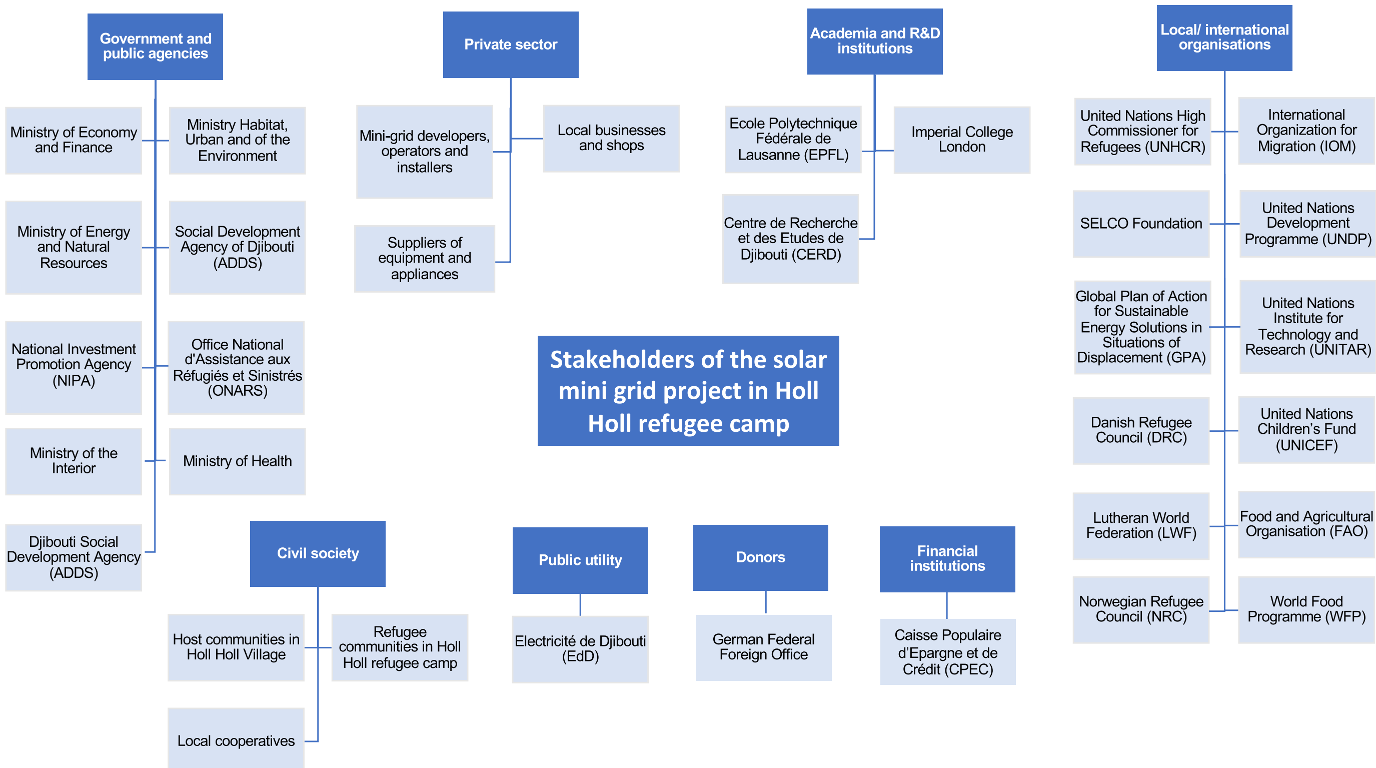

Fig. 3. A stakeholder map for the solar mini grid project in Holl Holl refugee camp.

important emphasis was made by the two main project leaders – the GPA CU/UNITAR and SELCO Foundation – on designing a context-specific intervention with an appropriate technology while also investigating possibilities to develop market-based solutions, that could ultimately create jobs and enable skills transfer [91]. However, since no conceptual planning framework related to energy access was in place within the organisations involved in this study, no guidance was retrieved from it regarding optimal community integration.

While the COVID-19 pandemic prevented the authors from accessing Holl Holl refugee camp to engage with the displaced population and the host community directly, proxies were used to evaluate the *socio-cultural* context. For example, a close collaboration with the two main project leaders, combined with remote interviews of key informants from organisations active locally, allowed us to understand the local power dynamics, preferences for technologies and current behaviours towards energy within the community. This way, it enabled us to capture the current situation and the desired future changes in regard to energy access to the best of the authors' ability.

For the case of Holl Holl refugee camp, UNHCR's core goal is to coordinate services provided by the other UN partner organisations (IOM, UNICEF, WFP, FAO), while ONARS is the governmental body responsible for the day-to-day of the camp. For electricity supply in Djibouti, ADDS is in charge of the rural electrification plans such as the one that was funded by the World Bank to Holl Holl Village. Finally, the GPA Coordination Unit is in charge of coordinating the energy project in Djibouti refugee camps while SELCO Foundation is providing recommendations on how to build an ecosystem of stakeholders able to deliver sustainable energy and livelihood solutions. Table 2 below summarises the key stakeholders of the solar mini grid project in Holl Holl refugee camp.

This stakeholder mapping highlights how choosing to integrate an inclusive approach component in the planning and design phases allows us to engage on the questions related to the *social* and *cultural* aspects. The understanding of the *socio-cultural* context is key to take into account how local communities view their situations and desires for the future, and the ways in which energy can contribute to them [27]. While the depth of our needs assessment in this respect was limited by the restrictions on travel due to the COVID-19 pandemic, considering the roles of different stakeholders helped us to identify common pitfalls such as a lack of a sense of ownership which can often lead to reduced public acceptance and, subsequently, in the failure of the energy project [115].

5.2.2. Assessment of the desired impacts

The *Theory of Change* maps how and why the intervention is expected to achieve its stated long-term goals for the key stakeholders and shows a logical process of how the project activities lead to impacts (see Fig. C1 in Supplementary materials). In the case of Holl Holl, it was assessed that households could not be connected directly to the mini due to technical barriers, including the poor quality of the roof of the shelters and potential safety issues. Therefore, the solar mini grid is designed to power the humanitarian organisations' facilities, community facilities, streetlights and local refugee businesses. In doing so it will provide new and/or improved opportunities for livelihoods and resilience, benefits for the host population, better security and protection, as well economic and energy security, and quality education and health care services, as well as reduced impacts on the environment. Using the *Theory of Change* as a blueprint, the *Impact Canvas* (Fig. 4) maps the logical sequence of the intervention from the current situation to its impacts, enabling us to identify the steps that need to be taken to successfully implement the project.

For the intervention in Holl Holl refugee camp, the main inputs are initial funding for the intervention to sustainably run over the long-term, human capital to install and operate and maintain (O&M) the electricity system, spare parts of the mini grid, and efficient appliances and equipment. Activities required to deliver the intended impacts are the procurement and installation of the mini grid, and the training of local staff on the use and maintenance of the equipment for the humanitarian organisation and community facilities and the business appliances. Additional activities will be required to ensure the effective usage of the energy services. The connection of business owners to local

Table 2
Summary of key stakeholders in the solar mini grid project in Holl Holl refugee camp.

Key stakeholders	Mandate and role in Djibouti and the project
Refugee and host communities in Holl Holl	Beneficiaries of the project, with needs and aspirations regarding the daily energy usage including productive uses of electricity (e.g. local businesses within the camp and in the Village). Active stakeholders involved in decision-making and design of the energy intervention since they are ultimately one of the main users of the solar mini grid.
United Nations High Commissioner for Refugees (UNHCR)	Responsible for the protection of refugees, coordination of services for refugees provided by its UN partner organisations, and the contracting of NGOs providing humanitarian and development aid.
Office National d'Assistance aux Réfugiés et Sinistrés (ONARS)	State body set up to assist refugees to gain access to humanitarian and development aid and to issue refugee and residence permits during their stay in the country. Responsible for the day-to-day management of the camps and the distribution of food and non-food aid.
Djibouti Social Development Agency (ADDS)	Its mission is to contribute to the eradication of poverty among vulnerable groups and to reduce the disparity between regions. It is also in charge of the operations in the rural and peri-urban areas in order to electrify areas not covered by the Electricité de Djibouti (EdD) network.
Global Platform for Action on Sustainable Energy in Displacement Settings (GPA) Coordination Unit (CU), hosted at UNITAR	Responsible for the project planning and coordination, delivering the energy assessments and reports, and monitoring the progress.
SELCO Foundation	Identify an ecosystem of stakeholders who can collaborate towards delivering sustainable energy and livelihood solutions to the displaced populations and vulnerable communities of Djibouti and recommend programmes that can enable the delivery of the said solutions.

microfinance institutions, for example through a partnership with Caisse Populaire d'Epargne et de Crédit (CPEC) [113] could support the acquisition of machines for productive uses such as for refrigeration, food processing, and tailoring. Vocational and entrepreneurship training for displaced people (and their host community), meanwhile, would strengthen local capacities.

Examples of measurable outputs for this intervention are the number of facilities and businesses connected to the mini grid, the number of facility staff and beneficiaries having received training, the number of local entrepreneurs connected to formal (micro-)finance, the reliability of the electricity system (defined as the number of hours when electricity is available (i.e. without blackout) divided by the total number of hours of operation of the system over its lifetime) and the number of hours of street lighting at night (see non-exhaustive list of indicators suggested in Fig. 4). Ultimately, there are a diverse range of outcomes related to socio-economic development, reduced environmental burdens, improving the enabling environment as well as the quality of life of the refugee and host communities. These outcomes translate into concrete impacts such as increased number of refugees completing primary school, accessing primary health care, while having higher disposable income and lower unemployment rate. Additional results are better social inclusion of refugees within the host community as well as climate change mitigation resulting from reduced GHG emissions.

Specifying the desired long-term impacts using this canvas is a way to capture the *social-cultural*, *economic* and *environmental* changes that the *technological* intervention is meant to bring about. Moreover, it is an opportunity to reflect on the potential unintended impacts at all the levels that could become negative externalities – and how to mitigate them. An example would be an unfair access to the new improved services for marginalised groups in the camp. On the other hand, the inputs and activities are setting the prerequisites for success and the sustainability of the intervention delivery model, which will ultimately influence the design of the *business model*. As a complement to the *Impact Canvas*, the *EDM Toolkit* highlights the importance of the supporting services (which focus on weaknesses or gaps in the *political-legal* enabling environment or *socio-cultural* factors) that are of key consideration for the energy service to be delivered in a sustainable manner [92]. Identifying these complementary activities is critical to reduce the risk of planning and designing a project that will not be sustained over time after the intervention of the humanitarian and/or development actors. Finally, the *Impact Canvas* creates a foundation for evaluation, programme development, organisational change, and strategic planning. Moreover, it is a useful tool to communicate more effectively with internal and external stakeholders.

5.2.3. Energy system modelling

The next step of the framework is to refine the *technological* solution and its technical performance: we do this by using energy system modelling to identify an optimal electricity system that meets the needs of the end-users. Based on the assessment of the *socio-cultural* context and the stakeholders mapping, the energy intervention is then narrowed down to a context-specific solution.

The optimisation of the mini grid was performed under the condition of providing electricity at least 95% of the time, with the optimum system being that with the lowest cumulative costs over its lifetime. The cumulative cost of a system, or total cost of a system over its lifetime is equivalent to its Net Present Cost (NPC) over its lifetime with the corresponding discount rate (i.e. 5% discount rate in that case [118]). In other words, it is the sum of all system costs, including installation, equipment, fuel, and operation and maintenance costs over the lifetime of the system. In the case of Holl Holl refugee camp the electricity demand had to be estimated to provide a load profile as an input to the energy modelling software. It was chosen to follow the method suggested by Sacino et al. [61], Sacino [62] and Ziade [63] (see Section B in Supplementary information for further details). Thus, we computed the electricity demand based on the usage of electrical appliances and equipment that would enable the desired changes, identified in the previous Sub-section, to take place. For instance, we considered the usage and power requirements of basic health care equipment and digital education materials, and machines and appliances for productive uses. Therefore, the appliances and equipment referred to as inputs in the *Impact Canvas* are directly reflected in the modelling of the electricity loads.

Using the *CLOVER* software, we found that to meet an electricity demand of 65.8 kWh/day, the optimal system is composed of 22 kWp of solar PV and 75 kWh of battery storage capacity. Overall, the total costs (including installation and transport) were estimated to be USD 88,717 over the course of 12 years of operation and result in an average levelised cost of used electricity (LCUE) of USD 0.313/kWh. These outputs from the modelling are key as they directly influence the design of a sustainable *business model* and are subsequently used to compute the *Profit and Loss Forecast*. Regarding the environmental impacts, it was estimated that 88.75 t of CO_{2eq} would be associated with the implementation and use of the system over 12 years (such as from the manufacture of the components, the operation of the system and the fuel used, among other aspects), with an emissions intensity of 274 gCO_{2eq}/kWh (see Section D in Supplementary materials for more details).

While this is the optimal size of solar PV-battery system to satisfy this demand, it is still important at this point to compare the solar mini grid solution with alternative electricity systems (in this case a grid extension and a diesel generator) in terms of *economic* and *environmental*

Situation	Inputs	Activities	Outputs	Outcomes	Impacts
• 47% attendance at school [ref. No.1] • No teaching materials in classrooms • Limited access to primary health care services • Very limited electricity access for productive uses • Limited activities after dark due to lack of streetlights • High unemployment rate (73% for women and 82% for men - [ref. No.2]) • High dependency on food rations and cash support • Lack of (perceived) safety	• Funding scheme for investment costs and O&M of mini-grid • Human capital for training, O&M and installation of the mini-grid • Mini-grid system • Energy efficient appliances • ICT equipment • Partnerships with microfinance institutions for businesses and households	• Procurement, installation and commissioning of the mini-grid • Training of refugee entrepreneurs and facility staff about usage and O&M of equipment and appliances • Entrepreneurship and vocational training • Post-training linkages • Distribution of start-up kits • Connection of refugee entrepreneurs to microfinance institutions • Facilitation of market linkages with nearest towns	• Reliable electricity access from the mini-grid is in place for basic services such as health, education and livelihoods • Streetlight is available at night for 12h • Refugee entrepreneurs, school, clinic and humanitarian staff is trained about usage and O&M of equipment and appliances • Refugee entrepreneurs have received business entrepreneurship and vocational training • Access to microfinance is in place	• Children study longer at night • Refugees have access to life-saving/ emergency health care (e.g. childbirth at night) • Attraction and/or retention of healthcare and teaching staff • Refugees have access to (tele-) communication and entertainment technologies • Refugee feel safer at night • New businesses and new activities at night are established in the camp • Businesses can use electricity to generate additional income • The dependency from refugees on humanitarian assistance is reduced as they are more self-reliant and resilient • The disparities between refugees and the host community are reduced • Greenhouse gases emissions are reduced	• Increase in number of children completing primary school • Increase in number of refugees accessing primary healthcare • Increase in disposable income of households • Decrease in unemployment rate • Better social and economic inclusion within the host community • Climate change mitigation
			Examples of Potential Indicators to Track Progress to Impact		
			• Energy system reliability (%) • Average N. of hours of streetlighting per night • No. of refugee entrepreneurs having received training	• No. of additional hours spent studying by children • No. of life-saving/emergency health care interventions performed • No. of new types of businesses established	• % of children completing primary school • % of unemployment in refugee camp

Fig. 4. Impact Canvas of the implementation of a solar mini grid in Holl Holl refugee camp (reference No.1: [116]; reference No.2: [117]).

considerations (see Section D in Supplementary materials). In the case of Holl Holl, we found that the cumulative costs of a grid extension of 3 km are 2.6 times higher than a solar mini grid. In addition, the solar-battery system option is a more environmentally friendly option: over 12 years of operation, the emissions intensity is 2.4 times lower than a grid-connected alternative. Compared to a diesel generator, a solar mini grid would require an initial investment approximately 4 times larger terms of equipment. Yet, the running costs (operation and maintenance) are more than 7.2 times lower and the cumulative costs over 12 years would be 1.5 lower while the emissions intensity is 4.6 times lower than a fossil-fuel based system. Overall, this comparison confirms that the solar mini grid is a sustainable and cost-effective alternative to the grid extension and a diesel generator.

However, we found that for some refugee entrepreneurs in the camp with low electricity consumption (20 Wh/day) buying a (pico-)solar home system (SHS) for USD 28.1 would be more advantageous. These systems would be more economical after 8 months, rather than paying a monthly bill for the mini grid connection at the national grid tariff. Entrepreneurs with greater energy needs would not be able to rely on a SHS, however, owing to the technological limitations of these smaller systems.

5.2.4. *Business model design*

Designing a sustainable business model requires an understanding of the enabling environment (such as the laws, policies, standards, infrastructure, global trends, institutions), as highlighted in the *EDM Toolkit* [92]. Since it is usually beyond the direct control of the project managers and beneficiaries, the enabling environment necessitates a careful analysis to determine what is feasible or not according to the frameworks already in place. In the case of the off-grid solar mini grid in Djibouti, several barriers and limitations were of great influence on the choice of ownership, procurement, and business models. For instance, the current laws in Djibouti only allow the public utility, Electricité de Djibouti, to charge consumers for electricity in this situation of monopoly [120]: a concession from the government is therefore required to sell electricity to customers and the tariffs need to align to the grid tariffs levels, if no subsidies are arranged. Therefore, the pros and cons of each type of mini grid ownership and procurement models were weighed by using guidance from the literature. In addition, recommendations and best practices from the literature specific to displacement settings were used to inform the choice.

The Moving Energy Initiative and Kube Energy identified three main pathways for the institutional transition to solar power for organisations operating in displacement settings: the purchase of solar assets (full ownership); a lease-to-own model; and through a power purchase agreement (PPA) [70]. From the humanitarian organisations' perspective, "energy-as-a-service" through a PPA would enable them to fully delegate the electricity production needed to power their local operations in Holl Holl. This would translate into the externalisation of the operation and maintenance (O&M, for which the humanitarian organisations have generally limited technical and human resource capacity [69]) and the billing process (which would help overcome the issue that humanitarian organisations are not entitled to send invoices to business owners as they are not permitted to receive money from refugees for such services). Furthermore, this model does not require any upfront capital cost (as would be the case under full ownership) and the monthly energy bills are usually lower than with a lease-to-own contract since it does not account for interest charges to finance the mini grid's purchase over the years [70]. Moreover, the time commitment of a PPA is typically more flexible and lower (3–5 years) compared to a purchase or lease-to own (5–25 years) models [70]. In addition, under a PPA the organisation does not hold a large and unmovable asset nor bear the risk associated with the responsibility of the O&M, as is the case for the other options. Finally, A PPA is favourable for the mini grid developer since it decreases the risks incurred by ensuring fixed revenues from the anchor customers. Based on the considerations found in the literature, we concluded that the most suitable humanitarian procurement model for electricity would be a PPA, used in conjunction with a private sector ownership model where a company would own the energy system and be responsible for its O&M.

For the case of Holl Holl, it was found that a private sector model would be the most suitable delivery model. The key reasons are the lack of interest from the public utility for small-scale mini grids and the current limited technical capacity of the refugees to operate the system themselves. While the enabling environment, including the regulations and policies, are not yet in place to fully support independent power producers and mini grid developers in Djibouti, the private model seems to be the most suitable one to deliver quality energy services to both refugees and humanitarian organisations in Holl Holl refugee camp. This recommendation is aligned with the current advocacy movement in the humanitarian energy sector to engage with private actors and design market-based solution for sustainable energy solutions [15]. From the perspective of the refugees and their host community, the proposed model is not only delivering the energy services identified as a need in the first place, but it might also enable economic development, social inclusion, and jobs creation (such as for the O&M of the mini grid in case the private owner considers it beneficial to train several local people) in Holl Holl refugee camp, and potentially in Holl Holl Village too (where the host community resides).

Despite all the advantages listed, some caution should still be exercised. Selecting the right private sector partners to participate in an innovative procurement, ownership and/or financing model necessitates a well-designed application process, avoiding unnecessary hurdles to participation, and performing due diligence before contracting [121]. Furthermore, the negotiation of a PPA with a third party might be complex and contracting arrangement costs might turn out to be more expensive than traditional in-house procurement due to the novelty of the model in the Djiboutian humanitarian context. In addition, the current regulatory framework in Djibouti only allows the public utility (EdD) to sell electricity to consumers, which means a concession would be necessary. To counter these problems, one suggestion would be to follow the recommended standard set of contractual terms for PPAs, which were recently drafted in other work undertaken by the GPA Coordination Unit [122].

Once the enabling environment is understood and the ownership and procurement models are defined, it is possible to develop a *business model* by using the *Social Business Model Canvas*, as we present in Fig. 5. The first area to highlight are the parts that make the enterprise “social” compared to the “classic” one from Osterwalder et al. [86], i.e. the targeted impact, the mission and values, and the beneficiary needs. In the present case, it would be an impact-driven company whose mission is to provide clean energy to its customers, while maximising local socio-economic development and generating sustainable revenues from its operations. The target impact and beneficiary needs have already been defined in the *Impact Canvas* in Fig. 4. Furthermore, the core of the *Social Business Model Canvas* is the value proposition suggested to the customers, in this case the governmental institutions (ministries in charge of the education and health), the humanitarian organisations (UNHCR, NRC, LWF, etc.) and the refugee business owners. The common value offered by the company is reliable and affordable electricity as a service, from which other attractive benefits ultimately arise. The key activities, resources and partner network highlight the needs for capacity building, awareness raising, and external support from the government and organisations to reinforce the ecosystem around the business. Finally, the cost/benefit balance suggests that the *economic*, *environmental*, and *social* benefits would outweigh the costs.

All in all, Fig. 5 highlights that the *business model* aspect is clearly the key aspect in the canvas, while the *social-cultural* and *environmental* aspects are captured by the values, missions, targeted impacts and beneficiary needs, as well as the *social* and *environmental* costs and benefits. The *political-legal* aspect is reflected in the ownerships and humanitarian procurements models, and in the partner network, key activities and key resources. The *economic* aspect is considered in the financial costs, revenue streams and cost-benefit analysis. Finally, the *technological* aspect is not explicitly considered in the canvas but indirectly appears through the value proposition with the reliability and cleanliness.

5.2.5. Economic viability assessment

The last component of our framework is the *Profit and Loss Forecast*, wherein we apply simplified tariff and financial models to the solar mini

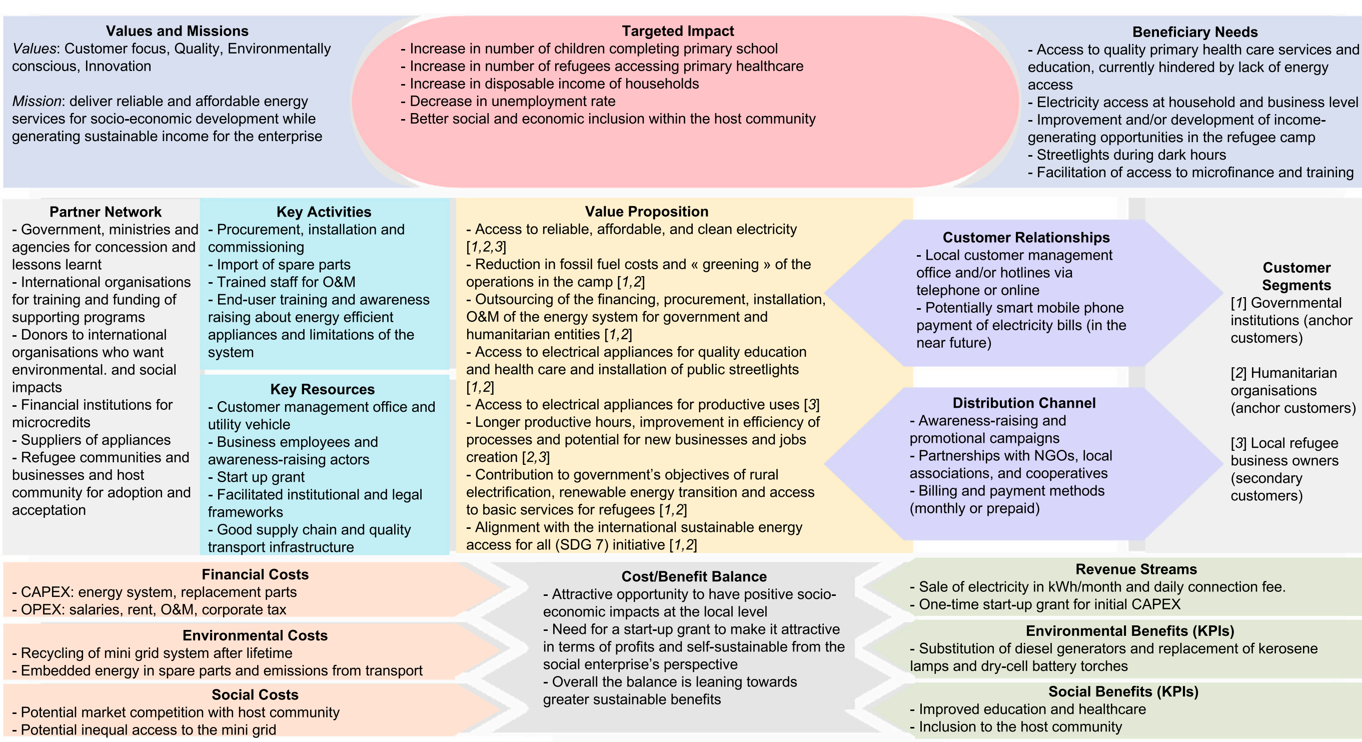

Fig. 5. The Social Business Model Canvas for the solar mini grid in Holl Holl refugee camp. Note: The numerals in the brackets under the Value Proposition column refer to the respective Customer Segments that could be served by that Value Proposition.

grid ownership, procurement and *business* models defined earlier. For the case of Holl Holl, the capital expenditures (CAPEX) were based on the financial results from CLOVER. The installation and logistics costs were estimated to be 20% of the CAPEX, while the operational expenditures (OPEX) were computed via an estimation of the O&M costs per unit capacity of the solar and battery storage (see Section E in Supplementary information for further details). The tariff levels and daily connection fee were assumed to be the same as the national tariffs set by Electricité de Djibouti [107] for compatibility with the enabling environment. Finally, it was usually assumed that an Internal Rate of Return (IRR) – a common measure to evaluate projects as it represents the return achieved by an investment and is sometimes viewed as a measure of efficiency [123] – of 20% or more on capital investment would be attractive for investors [103].

Using these key financial inputs, we calculated that reaching an IRR of 20% would require a 78% start-up grant for the initial investment costs, leaving a 22% equity for the company. The start-up grant (USD 48,364) could therefore cover the majority of the first-year CAPEX (USD 51,970) and of the installation and logistics costs (USD 10,394), with the remainder to be met by the company. Overall, the results of the *P&L Forecast* show that with a start-up grant, the enterprise owning, operating and maintaining the solar mini grid could be profitable after three years. On the other hand, without any grant funding it would take fifteen years to break even and therefore not be attractive for a private company in the energy sector.

The simplified *P&L Forecast* presented in Table 3 captures the aspect of *economic* attractiveness, while tying it back to the *business model* (See Section E in Supplementary materials for the detailed *P&L Forecast*). It also considers the *political-legal* environment of the country, with the defined tariff levels and the corporate tax. All in all, this last step of the framework is key to verify the sustainability of the energy intervention in terms of *economic* viability.

6. Discussion

6.1. Coverage, gaps and limitations of the presented framework

The framework we present in this paper allows us to systematically address the six aspects (*social-cultural*, *environmental*, *political-legal*, *business model*, *economic*, and *technological*) of sustainable energy interventions, often considered independently and overlooked in the planning and design phases. Utilising a variety of tools, both illustrative (*Theory of Change*, *Social Business Model Canvas* and *Impact Canvas*) and simplified (e.g. *P&L Forecast*), as part of a single central framework could make it more straightforward to ensure that each sustainability aspect is properly addressed and none are neglected. The framework aims to reduce the potential number and impacts of unforeseen project risks by identifying and managing them earlier in the project management cycle, as demonstrated in the case study of Holl Holl refugee camp. A strength of this approach also resides in the fact that it can be improved through iteration: as more information comes to light in later steps, or further through the design stages, it can be used to re-inform earlier processes as the overall understanding develops.

By combining project design tools, the framework helps to conceptualise a potential energy intervention in a systematic manner despite limitations related to the availability of information (in our case, using secondary data only). While data might be scarce, and its accuracy or relevancy to the project limited, the steps of this framework can help to reduce the uncertainty inherent in such projects. In particular it is often complicated or unfeasible to access refugee camps to undertake detailed surveys to collect data, and even more during the COVID-19 pandemic; using this approach it remains possible to establish an overview of the intervention, its impacts, and the opportunities for local development that can result from it. Furthermore, it is possible to critically evaluate the potential for a sustainable business to be established, including the financing, installation, operation and maintenance of the suggested electricity system in the long term.

Even after using this framework, a user may still face some unknowns and remaining uncertainties. For instance, many assumptions regarding the usage and choice of technologies for the solar panels (monocrystalline, polycrystalline, thin film, etc.) and the batteries (lead-acid, lithium-ion, etc.) might need to be further investigated based on which are selected for implementation. This is directly relevant to the energy system modelling, which is dependent on numerical parameters that are specified *ex-ante* based on reasonable assumptions but will change when particular technologies are selected for implementation. This will also impact the business model design, for example, as different possibilities can still be considered in the procurement processes later in the project cycle based on their availability on the market and the preferences of the future mini grid owner. Once these are defined it would be possible to iterate these two design steps, if necessary, to identify the relevance of any difference in technology choice.

One way to reduce the uncertainties linked to the energy system and *business model* would be to invest some time and resources to refine the understanding of the local energy needs, priorities and willingness to pay. This can be done by conducting surveys at the household, community and institutional facilities, and enterprise levels; if there are electricity supply technologies already in place, it would be recommended to install metering systems to monitor the current electricity usage. Another way would be to consult the stakeholders as a final step to make sure the intervention will meet their expectations, and it is feasible and suitable for the local context (for example compatible with the long-term plans for displacement settings set by the government). This last iteration can help ensure that the electricity system will be sustained if the humanitarian organisations were to exit the project.

It could be argued that the way the framework was presented implies that a single energy solution (i.e. one technology, such as a solar mini grid) can address all the energy needs identified. However, the authors would like to acknowledge that it is unlikely that a “one size fits all” solution is adapted for all the end-users in terms of socio-cultural acceptance, affordability or other considerations. Indeed, the comparison of the solar mini grid with pico-solar SHS in the case of Holl Holl shows that it is often the case that there is a need for a range of solutions: aside from the economic comparison between electrification options, some refugees may wish to have direct ownership and control over the assets which could support their livelihoods and help uplift them from poverty [119], in this case preferring these smaller SHSs rather than a connection to the mini grid. These findings highlight the importance of putting into perspective the affordability of different energy solutions by end-users and the impacts it has on the sustainability of the *business*

Table 3
Profit and Loss Forecast for the solar mini grid in Holl Holl over 12 years with a start-up grant of USD 48,364 covering 78% of the CAPEX.

Year		1	2	3	4	5	6	7	8	9	10	11	12
Costs	Total costs (USD/year)	15,670	1670	1670	1670	7513	1374	1374	1374	5214	1130	1130	1130
	Cumulative costs (USD)	15,670	17,340	19,010	20,680	28,193	29,566	30,940	32,314	37,528	38,659	39,789	40,919
Revenues	Total revenues (USD/year)	7972	7972	7972	7972	7972	7972	7972	7972	7972	7972	7972	7972
	Cumulative revenues (USD)	7972	15,943	23,915	31,886	39,858	47,829	55,801	63,772	71,744	79,716	87,687	95,659
Profits	Net profits after tax (USD/year)	4726	4726	4726	4726	4948	4948	4948	4948	5131	5131	5131	5131
	Cumulative profits (USD)	−9274	−4547	179	4905	3714	8662	13,611	18,559	19,606	24,737	29,868	34,999

model.

Very much like "fuel stacking" (the practice of using multiple combinations of stove and fuel for cooking within the same household [124]), the solutions used by households, business owners, and humanitarian organisations for electricity supply might vary. Furthermore, the electricity system presented in the case study does not aim to address all the significant energy needs of the community – for example the need for improved access to clean cooking solutions, a need for which electricity could offer an alternative solution – and so further energy interventions would be necessary to provide for these important aspects. Therefore, while the framework presented focuses on how to holistically approach a single sustainable energy intervention, it does not imply that one particular technology will respond to all the needs. The authors would encourage the planners, where possible, to keep these considerations in mind to avoid potentially resolving one issue without also addressing other fundamental needs.

6.2. Key take-aways from applying the framework to the case study

The theoretical case study of the solar mini grid in Holl Holl refugee camp demonstrates the five steps of the suggested framework as it moves forward the planning and design of the energy intervention from an uncertain starting point to a more well-developed project proposal. Initially, the intended activities and impacts of the desired energy intervention were undefined, and the influence of the local context (in terms of *economic*, *social*, *cultural*, *political*, and *legal* considerations) was unclear. Furthermore, the intervention was seen as a stand-alone project and supporting activities and services, such as microfinance and training, were not embedded within the initial scope. In addition, the electricity demand of institutional and community facilities located in the camp was not estimated, the *social* and *economic* benefits of connecting the refugee businesses were not quantified, and the *environmental* benefits compared to alternative systems were not scoped.

Applying this framework enabled us to shed light on the key aspects that could have an influence on the success of the potential implementation of a solar mini grid in Holl Holl refugee camp. We are able to articulate and logically map out the impacts and activities that should take place to deliver sustainable electricity services to the refugees and organisations in the camp. Our recommendations for the ownership, humanitarian procurement and *business* models consider the constraints and opportunities linked to the enabling environment and help to flag up potential issues and entry points for lobbying and advocacy.

Our case study of Holl Holl featured areas in which further research or investigation could have strengthened the overall planning of the proposed intervention. The technical modelling could be beneficially refined through greater confidence in the electricity demand profiles, and hence in the system designed to meet those needs: no specific and recent energy surveys had been conducted, which made these inputs less certain and less tailored to the local needs. We therefore recommend undertaking a detailed assessment (based on a field visit) of the electricity needs of the primary school, clinic, offices and refugee businesses and properly evaluate the demand based on the desired electrical appliances, which is key to validate the assumptions made in this study. By doing so, we expect this to capture the schedule and habits of the relevant actors in the camp and to better anticipate potential changes in behaviour due the energy intervention. Moreover, the survey should include the assessment of the economic capacities to purchase the equipment and the willingness-to-pay for the future consumers of the mini grid.

Furthermore, it would be necessary to consult with the organisations and groups identified in the stakeholder mapping (for example community representatives, UNHCR, ONARS and ADDS) to present the proposed intervention and gain their endorsement before scoping the next stages of the implementation. We therefore suggest conducting common discussions with stakeholders, including focus group discussions, in order to make the decision-making process as inclusive as possible. This could identify any potential issues with the intervention itself and, if these are resolved, set a foundation for procurement and the realisation of the intervention in Holl Holl.

Overall, our framework allowed us to systematically address all six sustainability aspects, rather than only a subset, in the planning and design phases and therefore could reduce the associated risks during implementation of the project. Finally, this case study highlights some of the complex challenges of considering each aspect in the design process (such as enabling environment, policies, *socio-cultural* context) but also the value in approaching intervention design in a holistic manner such as this.

6.3. Replication in other displacement settings, contexts, and types of energy interventions

The framework we present could be used in other types of displacement settings beyond rural refugee camps in the Horn of Africa. Displacement settings have different forms (formal camps, informal camps, settlements, etc.) and characteristics (rural, peri-urban, urban, etc.) with distinct needs and therefore require tailored energy solutions. Yet, our framework offers an approach to planning and designing energy interventions which is sufficiently flexible and holistic to accommodate the diversity of displacement settings and needs. The application of the framework would consider the same six core aspects, even though they might have differing levels of influence in each context: displaced people living in a rural refugee camp established in the 1990s, like Holl Holl, have different requirements compared to those living in an urban setting, where more than half of displaced people reside [125]. Furthermore, focusing on urban areas is key since, while energy access is a challenge in many rural areas, many urban populations also face a variety of issues which have not yet received sufficient attention [126,127].

Furthermore, even though this paper discusses a case study about a fully renewable and decentralised (off-grid) electricity system, the framework remains applicable for the extension of a centralised grid network. For instance, the solar mini grid for Holl Holl was compared to a grid extension and the same sustainability aspects were considered in the evaluation. Moreover, energy modelling tools such as *CLOVER* and *HOMER* allow the design of on-grid renewable electricity systems, when appropriate, and for settings when the grid is available nearby. Therefore, the value of centralised electricity systems and the importance of leveraging available energy resources are acknowledged in the approach presented in this article.

This approach could also be utilised for other types of energy interventions than solar mini grids, such as clean cooking solutions (e.g. LPG [128]). The energy modelling software can be substituted from *CLOVER* to any suitable tool which will help to assess the size and *technological* characteristics of the energy systems required, as well as the *economic* implications and *environmental* impacts. The other components of our framework will remain more or less unchanged, regardless of the types of displacement settings targeted or energy interventions envisioned, as they are more generally applicable from the outset.

Furthermore, as the basis for the framework was drawn from the *Energy Delivery Model Toolkit* from Garside and Wykes [92], whose scope is focused on poor and marginalised communities, we suggest that it can also serve to plan and design interventions for energy deprived areas that do not necessarily host displaced populations. While we included the *Embedded and Inclusive Programming Framework* for humanitarian energy services, from Rosenberg-Jansen et al. [91], it could similarly be transposed to non-humanitarian programmes and projects. In other words, our holistic approach could be replicated where there is a need to improve the access to energy services, regardless of the displacement status of the community.

Any replication of this framework, however, would rely on the involvement and support of multiple stakeholders with potentially diverse needs and objectives which could, for example, vary between displacement and non-displacement settings. The involvement of a project manager responsible for the coordination and management of the stakeholders and the project overall, or an oversight committee performing a similar role, could be a critical enabler in facilitating this. Our theoretical application of the framework to the case of Holl Holl would similarly require a dedicated management focal point if practical implementation of the mini-grid intervention were to take place.

6.4. Recommendations for the planning and design of sustainable energy interventions in displacement settings

In this paper we presented that taking a holistic and "interdisciplinary" (i.e. linking different disciplines into a coordinated and coherent whole [129]) approach, as suggested in our framework, could potentially reduce uncertainties and risks of failure. This entails that a range of expertise, to address all six aspects, need to be leveraged during the planning and design phase of an energy intervention in order to maximise its chance of success. While it would be ideal to have a project team composed of sociologists, anthropologists, businesspeople, engineers, and international development practitioners to cover each aspect and fully appreciate the others, this is rarely feasible. We therefore encourage "multidisciplinary" (i.e. drawing on knowledge from different disciplines but staying within their boundaries [129]) teamwork despite the complexity of the initiative.

The main problems identified by Montedonico et al. [130] during "interdisciplinary" work are the common understanding of fundamental concepts (such as sustainability), the domination of one discipline over the others and the different visions of the priorities within the same project. To overcome the associated challenges with this type of multiple disciplinary work, we also recommend that the application of our framework involves five useful principles for "transdisciplinary" (i.e. integrating the natural, social and health sciences in a humanities context, and transcending their traditional boundaries [129]) work related to community renewable energy projects. These, recommended by Thomas et al., are preparation, common understandings, flexibility, shared responsibility, respect, and lightheartedness and humour [131]. All in all, the framework we present can be used as a guide to ensure that no sustainability aspect is forgotten and, even if specific expertise is not available, its consideration is highlighted as an area for further development or potential uncertainty.

We recommended asking for insights from specialists from different disciplines to support the aspects with which the project team has less expertise. For example, a team with mostly technical backgrounds might have issues capturing the *political-legal* and *social-cultural* characteristics of the energy intervention. Furthermore taking a multiple disciplinary approach is recommended since energy access is such a cross-sectoral issue within the humanitarian sector [57], as it touches on many areas of the cluster system such as food security, shelter, camp coordination and management, water, hygiene and sanitation. One way to benefit from external support related to multiple disciplinary work would be to reach out to the Humanitarian Energy Exchange Network (HEEN), the central platform for coordination and collaboration between humanitarian and development agencies working on improving sustainable energy access of displacement-affected communities [132]. The HEEN supports a holistic approach to programming and serves as a technical resource to governments, non-governmental organisations, intergovernmental organisations, and practitioners, using lessons captured from the experience of network members [132].

Our approach suggests that most of the value is added only by addressing every aspect through the five steps of the framework. For example, if a project manager only considers a solution with the lowest costs (*economic*), they might overlook the environmental impacts of recycling the batteries (which are included in through the use of *CLOVER*) or the social-cultural acceptance (which could impact the willingness to pay of community, important for tariffs and potential subsidies). Neglecting one or more aspects could have significant impacts on the success of the intervention and therefore the project manager should acknowledge the weight of each aspect and its influence.

Using this framework necessitates an initial investment and focus on the planning and design phase which may be challenging to implement if a project is under pressure from resource constraints, organisational requirements, or a pressing need from the community themselves. Acknowledging this, we recommend that future projects integrate greater resources for intervention planning and design where possible, ideally with the endorsement of supportive stakeholders, to weigh up this investment against the added value and sustainability it can bring in the long term.

In addition to engaging in a holistic and multiple disciplinary approach, it is important for designers of sustainable energy interventions to learn from previous projects, and in particular their shortcomings, of both the humanitarian and energy sectors. While there are limitations specific to their mandates and to the local context, humanitarian organisations can also apply lessons learnt from private sector energy projects as well as those from development organisations. Our framework is not a way to reinvent the wheel on how to plan and design energy interventions in humanitarian settings but rather a suggestion on how to systematically approach them. For that reason, with this framework, we aim to encourage the incorporation of the learning from other projects and sectors by ensuring that none of the core sustainability aspects are neglected.

6.5. Further research

The case study of the solar mini grid in Holl Holl refugee camp has brought forward several points that warrant further research. First, the causal link between electrifying local businesses in displacement settings and the intended socio-economic impacts is not clear yet in the literature, and a better understanding of the enabling environment required for them to thrive could be gained through dedicated studies. We therefore recommend researching if, and how, electricity can be a means to give refugee entrepreneurs the opportunity to pursue their goals and increase their quality of life; implementation of a sustainable electricity system in Holl Holl could provide a representative case for such an investigation.

More generally, there is an opportunity for similar studies in social science research. For instance, there is a need to better understand the link between electricity access, sustainable livelihood and economic development opportunities, especially under the potential constraints imposed by displacement settings. As suggested by Terrapon-Pfaff et al. [133], research is required to understand what the key factors in the energy delivery model are and what external geographic, climatic or economic factors determine the achievement of development outcomes and impacts. These could be derived from specific case studies and, where possible, generalised across other humanitarian situations. Furthermore, research focusing on energy justice, for example how the benefits or drawbacks of energy systems are distributed throughout society and how fair and representative decision-making between stakeholder groups can be achieved when designing such systems [134], should be part of the future humanitarian energy research agenda. While this article does not extensively address how energy justice applies in displacement settings, the concept represents a valuable decision-making tool that can support energy planners and consumers to make more informed energy choices [135] and should therefore be further explored. In addition, Sovacool and Dworkin [135] suggest that energy justice implicates that access to energy services is a universal right irrespective of the level of economic development of its citizens. Thus, it would be pertinent to better understand how the concept can support policy makers to make the case for the inclusion of energy access as a basic right for displaced people a reality (for example through inclusive national energy and development frameworks and planning), which is currently overlooked by most host governments and humanitarian organisations.

The development of the presented framework brings to light new pathways for subsequent research. For example, we suggest that our approach is likely to reduce the risks of failure of an energy intervention in displacement settings as it considers the six aspects defined as critical for sustainability. This could be verified counterfactually via empirical case studies that do not specifically address the six aspects and identifying if it affected their success, and if so in what ways. Ideally these could be compared to case studies of energy interventions in which our framework is used, like the one of Holl Holl if implemented in the future. It would then be potentially possible to draw generalisations between the cases and validate the value of our approach with evidence-based arguments.

Finally, we argue that our framework has been designed to be put into practice and used by the humanitarian energy sector. Even though it has been developed using an academic approach, the goal is to encourage its adoption by the humanitarian actors as a useful tool for practitioners. One way to do so would be to embed this into existing online training courses, such as those developed by UNITAR and WFP on Energy Delivery Models [136]. Furthermore, as the number of energy experts deployed within humanitarian organisations grows (e.g. through NORCAP, the Norwegian Refugee Council's (NRC) global provider of expertise to the humanitarian, development and peace-building sectors), there is a potential to build capacity and encourage the integration of the framework in the best practices for the humanitarian energy sector.

7. Conclusion

In this paper we have presented a new holistic framework that promotes the systematic consideration of all of the identified key sustainability aspects of energy interventions, namely the *social-cultural*, *environmental*, *political-legal*, *business model*, *economic* and *technological* aspects. We explored Holl Holl refugee camp as a theoretical case study example to demonstrate how our framework can support the planning and design of sustainable energy interventions in displacement settings. Furthermore, we have highlighted how existing planning and design tools can be combined to form synergies, in order to address the sustainability challenges linked to energy systems planning and design in displacement contexts. In addition, we have suggested that a coherent and comprehensive approach – such as the framework we presented – can potentially contribute to reducing the risk of overlooking important factors which might compromise the sustainability, implementation, and adoption of these energy interventions.

This framework provides a structured approach to support humanitarian practitioners, governments, and their partners in moving towards more holistic planning and design. Moreover, it challenges the current status quo such as traditional sectoral and short-term thinking that are characteristic to the humanitarian sector. The systematic approach of this framework is relevant to contexts other than displacement situations and electrification projects, but more broadly to any energy access intervention and its planning and design process, and can help to enhance sustainability practices in the humanitarian and development sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

[1] S. Pachauri, et al., Energy access for development, in: Global Energy Assessment - Toward a Sustainable Future, Cambridge University Press, Cambridge (UK) and New York (USA), 2012, pp. 1401–1458.
[2] F. Fuso Nerini, et al., Mapping synergies and trade-offs between energy and the sustainable development goals, Nat. Energy 3 (1) (2018) 10–15.
[3] IEA, SDG7: data and projections, International Energy Agency, Oct. 2020. https://www.iea.org/reports/sdg7-data-and-projections. (Accessed 2 March 2021).
[4] B. Tarekegne, Just electrification: imagining the justice dimensions of energy access and addressing energy poverty, Energy Res. Soc. Sci. 70 (2020), 101639, https://doi.org/10.1016/j.erss.2020.101639.
[5] R. Hiremath, S. Shikha, N. Ravindranath, Decentralized energy planning; modeling and application—a review, Renew. Sustain. Energy Rev. 11 (5) (2007) 729–752.
[6] P.A. Trotter, M.C. McManus, R. Maconachie, Electricity planning and implementation in sub-Saharan Africa: a systematic review, Renew. Sustain. Energy Rev. 74 (Jul. 2017) 1189–1209, https://doi.org/10.1016/j.rser.2017.03.001.
[7] M.J. Herington, E. van de Fliert, S. Smart, C. Greig, P.A. Lant, Rural energy planning remains out-of-step with contemporary paradigms of energy access and development, Renew. Sustain. Energy Rev. 67 (Jan. 2017) 1412–1419, https://doi.org/10.1016/j.rser.2016.09.103.
[8] A. Escobar, Encountering Development: The Making And Unmaking of the Third World, Princeton University Press, 1995.
[9] T. Mitchell, Rule of Experts: Egypt, Techno-politics, Modernity, Univ of California Press, 2002.
[10] J.C. Scott, Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed, Yale University Press, New Haven, Connecticut, 1998.
[11] L.H. Malkki, Speechless emissaries: refugees, humanitarianism, and dehistoricization, Cult. Anthropol. 11 (3) (1996) 377–404, https://doi.org/10.1525/can.1996.11.3.02a00050.
[12] G. Lahn, O. Grafham, Heat, Light, And Power for Refugees: Saving Lives, Reducing Costs, Chatham House, The Royal Institute of International Affairs, London, 2015.
[13] World Bank, Forced displacement: refugees, internally displaced and host communities, World Bank, Oct. 15, 2021. https://www.worldbank.org/en/topic/forced-displacement#1. (Accessed 15 November 2021).
[14] M. Bhatia, N. Angelou, Beyond Connections Energy Access Redefined, World Bank, Washington DC, 2015. Accessed: Feb. 24, 2020. [Online]. Available: https://www.worldbank.org/content/dam/Worldbank/Topics/Energy%20and%20Extract/Beyond_Connections_Energy_Access_Redefined_Exec_ESMAP_2015.pdf.
[15] S. Huber, E. Mach, Policies for increased sustainable energy access in displacement settings, Nat. Energy 4 (12) (Dec. 2019) 1000–1002, https://doi.org/10.1038/s41560-019-0520-1.
[16] R.L. Miller, M.A. Ulfstjerne, Trees, tensions, and transactional communities: problematizing frameworks for energy poverty alleviation in the rhino camp refugee settlement, Uganda, Energy Res. Soc. Sci. 63 (May 2020), 101404, https://doi.org/10.1016/j.erss.2019.101404.
[17] UNITAR, The Global Plan of Action for Sustainable Energy Solutions in Situations of Displacement: Framework for Action, United Nations Institute for Training And Research and Global Plan of Action (GPA), Geneva, 2018.
[18] O. Grafham, P. Sandwell, Harness better data to improve provision of humanitarian energy, Nat. Energy 4 (12) (Dec. 2019) 993–996, https://doi.org/10.1038/s41560-019-0518-8.
[19] C.B. Field, V.R. Barros (Eds.), Climate Change 2014: Impacts, Adaptation, And Vulnerability: Working Group II Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, New York, NY, 2014.

[20] L.Van Landeghem, Private-sector Engagement: The Key to Efficient, Effective Energy Access for Refugees, Chatham House, The Royal Institute of International Affairs, London, May 2016. Toolkit.

[21] R. Bellanca, Sustainable Energy Provision Among Displaced Populations: Policy And Practice, Moving Energy Initiative, London, 2014.

[22] J.W. Tester, E.M. Drake, M.J. Driscoll, M.W. Golay, W.A. Peters, Sustainable Energy: Choosing Among Options, MIT press, 2012.

[23] R.B. Nepal, Auditing project sustainability, Int. J. Gov. Audit. 21 (2) (1994) 10.

[24] S.C. Bhattacharyya, Energy access programmes and sustainable development: a critical review and analysis, Energy Sustain. Dev. 16 (3) (Sep. 2012) 260–271, https://doi.org/10.1016/j.esd.2012.05.002.

[25] B.K. Sovacool, Design principles for renewable energy programs in developing countries, Energy Environ. Sci. 5 (11) (2012) 9157, https://doi.org/10.1039/c2ee22468b.

[26] C.A. Miller, C. Altamirano-Allende, N. Johnson, M. Agyemang, The social value of mid-scale energy in Africa: redefining value and redesigning energy to reduce poverty, Energy Res. Soc. Sci. 5 (2015) 67–69, https://doi.org/10.1016/j.erss.2014.12.013.

[27] J. Cloke, Imagining renewable energy: towards a Social Energy Systems approach to community renewable energy projects in the Global South, Social Science (2017) 10.

[28] J. Alfaro, S. Miller, Satisfying the rural residential demand in Liberia with decentralized renewable energy schemes, Renew. Sustain. Energy Rev. 30 (Feb. 2014) 903–911, https://doi.org/10.1016/j.rser.2013.11.017.

[29] J. Watson, et al., What are the major barriers to increased use of modern energy services among the world's poorest people, and are interventions to overcome these effective? [Online]. Available CEE Review - Collaboration for Environmental Evidence, 2012, www.environmentalevidence.org/SR11004.html.

[30] D. Akinyele, J. Belikov, Y. Levron, Challenges of microgrids in remote communities: a STEEP model application, Energies 11 (2) (Feb. 2018) 432, https://doi.org/10.3390/en11020432.

[31] A.-L. Toba, M. Seck, Modeling social, economic, technical & environmental components in an energy system, Procedia Comput. Sci. 95 (2016) 400–407, https://doi.org/10.1016/j.procs.2016.09.353.

[32] R. Wüstenhagen, J. Boehnke, Business models for sustainable energy, in: Perspectives on Radical Changes to Sustainable Consumption And Production (SCP), 2006, p. 253. Copenhagen.

[33] A. Chaurey, P.R. Krithika, D. Palit, S. Rakesh, B.K. Sovacool, New partnerships and business models for facilitating energy access, Energy Policy 47 (Jun. 2012) 48–55, https://doi.org/10.1016/j.enpol.2012.03.031.

[34] E. Martinot, A. Chaurey, D. Lew, J.R. Moreira, N. Wamukonya, Renewable energy markets in developing countries, Annu. Rev. Energy Environ. 27 (1) (2002) 309–348.

[35] P. Ortega-Arriaga, O. Babacan, J. Nelson, A. Gambhir, Grid versus off-grid electricity access options: a review on the economic and environmental impacts, Renew. Sustain. Energy Rev. 143 (Jun. 2021), 110864, https://doi.org/10.1016/j.rser.2021.110864.

[36] T. Levin, V.M. Thomas, Can developing countries leapfrog the centralized electrification paradigm? Energy Sustain. Dev. 31 (Apr. 2016) 97–107, https://doi.org/10.1016/j.esd.2015.12.005.

[37] D.P. Kaundinya, P. Balachandra, N.H. Ravindranath, Grid-connected versus stand-alone energy systems for decentralized power—a review of literature, Renew. Sustain. Energy Rev. 13 (8) (Oct. 2009) 2041–2050, https://doi.org/10.1016/j.rser.2009.02.002.

[38] J.Baranda Alonso, P. Sandwell, J. Nelson, The potential for solar-diesel hybrid mini-grids in refugee camps: a case study of Nyabiheke camp, Rwanda, Sustain. Energy Technol. Assess. 44 (Apr. 2021) 101095, https://doi.org/10.1016/j.seta.2021.101095.

[39] ITDG, 'Household Energy in Emergency Situations', Boiling Point, no. 37, 1996.

[40] UNHCR, Global Trends Forced Displacement in 2020, UNHCR, 2021. Accessed: Feb. 24, 2020. [Online]. https://www.unhcr.org/60b638e37/unhcr-global-trends-2020.

[41] iDMC, Global report on internal displacement 2021 [Online]. Available, iDMC, 2021, https://www.internal-displacement.org/sites/default/files/publications/documents/grid2021_idmc.pdf.

[42] R. Muggah, A.E. Abdenur, Refugees and the city: the twenty-first-century front line [Online]. Available, Centre for International Governance Innovation, 2018, https://www.cigionline.org/sites/default/files/documents/WRC%20Research%20Paper%20no.2.pdf.

[43] J. Lehne, W. Blyth, G. Lahn, M. Bazilian, O. Grafham, Energy services for refugees and displaced people, Energy Strat. Rev. 13–14 (Nov. 2016) 134–146, https://doi.org/10.1016/j.esr.2016.08.008.

[44] S. Okiror, Massive deforestation by refugees in Uganda sparks clashes with local people, The Guardian, Feb. 18, 2019. https://amp.theguardian.com/global-development/2019/feb/18/massive-deforestation-by-refugees-in-uganda-sparks-clashes-with-local-people. (Accessed 12 November 2021).

[45] A. Martin, Environmental conflict between refugee and host communities, J. Peace Res. 42 (3) (May 2005) 329–346, https://doi.org/10.1177/0022343305052015.

[46] Moving Energy Initiative, The Moving Energy Initiative: innovating to improve sustainable energy access for displaced populations, Chatham House, 2015. https://www.chathamhouse.org/sites/files/chathamhouse/field/field_document/20150501MEIflyer.pdf.

[47] UNHCR, Global strategy for sustainable energy: a UNHCR strategy 2019-2024 [Online]. Available, United Nations High Commissioner for Refugees, 2019, https://www.unhcr.org/5db16a4a4.pdf.

[48] UNHCR, Clean energy challenge, UNHCR, 2021. https://www.unhcr.org/clean-energy-challenge.html. (Accessed 15 November 2021).

[49] Shell, Access to more: creating energy choices for refugees, Shell, Jul. 2020. https://www.shell.com/sustainability/communities/access-to-energy/access-to-energy-investing-in-communities/_jcr_content/par/textimage_copy_283397719.stream/1595953271615/db4946328e61328b1001fd5534929aa47dfe3db6/shell-enter-energy.pdf. (Accessed 15 November 2021).

[50] GPA, Why energy is important in displacement settings?, in: GPA - Global Platform For Action for Sustainable Energy in Displacement Settings UNITAR, Geneva Switzerland, 2021 [Online]. Available: https://www.humanitarianenergy.org/what-is-the-gpa/why-energy-is-important/.

[51] R. Gunning, The Current State of Sustainable Energy Provision for Displaced Populations: An Analysis, Chatham House,The Royal Institute of International Affairs, London, 2014. Research Paper.

[52] A. Boodhna, C. Sissons, J. Fullwood-Thomas, A systems thinking approach for energy markets in fragile places, Nat. Energy 4 (12) (Dec. 2019) 997–999, https://doi.org/10.1038/s41560-019-0519-7.

[53] GPA Coordination Unit, Blended Finance Solutions for Clean Energy in Humanitarian And Displacement Settings: Lessons Learnt – An Initial Overview [Online]. Available, UNITAR, Geneva, Switzerland, Jan. 2022, https://www.humanitarianenergy.org/news/latest/blended-finance-solutions-bridging-the-funding-gap-and-driving-self-sustaining-solutions-in-displacement-settings.

[54] S. Rosenberg-Jansen, T. Tunge, T. Kayumba, Inclusive energy solutions in refugee camps, Nat. Energy 4 (12) (Dec. 2019) 990–992, https://doi.org/10.1038/s41560-019-0516-x.

[55] Global Platform for Action on Sustainable Energy in Displacement Settings, Fossil fuel powered generators in humanitarian operations produce high CO2 emissions, Global Platform for Action on Sustainable Energy in Displacement Settings, Sep. 08, 2021. https://www.humanitarianenergy.org/news/latest/fossil-fuel-powered-generators-in-humanitarian-operations-causes-high-co2-emissions. (Accessed 11 November 2021).

[56] S. Rosenberg-Jansen, Refugee Energy 8, Oxford Refugee Studies Center, Oxford, UK, 2018.

[57] P.J.M. Thomas, P. Sandwell, S.J. Williamson, P.W. Harper, A PESTLE analysis of solar home systems in refugee camps in Rwanda, Renew. Sustain. Energy Rev. 143 (Jun. 2021), 110872, https://doi.org/10.1016/j.rser.2021.110872.

[58] D. Neves, P. Baptista, J.M. Pires, Sustainable and inclusive energy solutions in refugee camps: developing a modelling approach for energy demand and alternative renewable power supply, J. Clean. Prod. 298 (2021), 126745, https://doi.org/10.1016/j.jclepro.2021.126745.

[59] T. Ali, A.J. Nahian, H. Ma, A hybrid multi-criteria decision-making approach to solve renewable energy technology selection problem for Rohingya refugees in Bangladesh, J. Clean. Prod. 273 (2020), 122967, https://doi.org/10.1016/j.jclepro.2020.122967.

[60] D. Kirikkaleli, N. Doğan, Energy consumption and refugee migration in Turkey, Util. Policy 68 (2021), 101144, https://doi.org/10.1016/j.jup.2020.101144.

[61] J. Sacino, T. Fernando, S. Fani, Estimating the energy demand of refugee camps, in: CEED Seminar Proceedings 2017, 2017, p. 6.

[62] J. Sacino, Temporary off Grid Power Generation for Refugee Camps, Master thesis, The University of Western Australia, Crawley, Australia, 2017.

[63] H. Ziade, Renewable Energy Breakthrough Into Refugee Camps, Master Thesis, Ecole Polytechnique Fédérale de Lausanne, Lausanne, 2013.

[64] M. Franz, N. Peterschmidt, M. Rohrer, B. Kondev, Mini-grid Policy Toolkit: policy and business frameworks for successful mini-grid roll-outs, European Union Energy Initiative Partnership Dialogue Facility (EUEI PDF), Eschborn, Germany, 2014. https://www.ren21.net/Portals/0/documents/Resources/MGT/MinigridPolicyToolkit_Sep2014_EN.pdf.

[65] B. Tenenbaum, C. Greacen, T. Siyambalapitiya, J. Knuckles, From the Bottom Up: How Small Power Producers And Mini-grids Can Deliver Electrification And Renewable Energy in Africa, World Bank, Washington DC, 2014.

[66] M. Wiemann, S. Rolland, G. Glania, Hybrid Mini-grids for Rural Electrification: Lessons Learned, Alliance for Rural Electrification (ARE) and USAID, Brussels, 2014.

[67] GVEP International, The History of Mini-grid Development in Developing Countries, Global Village Energy Partnership, London, 2011. Policy briefing.

[68] USAID, What are the pros and cons of each ownership model, USAID, Feb. 13, 2018. https://www.usaid.gov/energy/mini-grids/ownership/considerations/. (Accessed 29 June 2020).

[69] IRENA, Renewables for refugee settlements: sustainable energy access in humanitarian situations [Online]. Available, International Renewable Energy Agency, Abu Dhabi, 2019, https://www.irena.org/publications/2019/Dec/Renewable-solutions-for-refugee-settlements.

[70] Moving Energy Initiative and Kube Energy, The Solar Energy Handbook: A Guide to Institutional Solar for Organizations Working in Humanitarian Settings, Chatham House and Kube Energy, London and Nairobi, 2018 [Online]. Available: https://mei.chathamhouse.org/file/2469/download?token=DFa3HvKW.

[71] Kube Energy, Our solar solutions, Kube Energy, Oct. 20, 2018. http://www.kubeenergy.com/solar-solutions. (Accessed 1 July 2020).

[72] I. Ben-David, T. Raz, An integrated approach for risk response development in project planning, J. Oper. Res. Soc. 52 (1) (Jan. 2001), https://doi.org/10.1057/palgrave.jors.2601029.

[73] V. Gambino, R.D. Citto, P. Cherubini, C. Tacconelli, A. Micangeli, R. Giglioli, Methodology for the energy need assessment to effectively design and deploy

mini-grids for rural electrification, Energies 12 (3) (Feb. 2019) 574, https://doi.org/10.3390/en12030574.
[74] E. Verploegen, Energy Assessment Toolkit Introduction: a community focused approach for identifying energy needs and market opportunities [Online]. Available, MIT D-Lab, 2017, https://dspace.mit.edu/bitstream/handle/1721.1/113313/Energy%20Assessment%20Toolkit%20Introduction%20%28v1.02%29.pdf?sequence=1. (Accessed 25 May 2021).
[75] SELCO Foundation, Profiling and needs assessment toolkit [Online]. Available, SELCO Foundation, 2014, https://selcofoundation.org/wp-content/uploads/2017/04/Need-Assessment-Toolkit-.pdf. (Accessed 21 May 2021).
[76] J. Mayers, S. Vermeulen, Stakeholder influence mapping [Online]. Available, International Institute for Environment and Development, 2005, https://policy-powertools.org/Tools/Understanding/docs/stakeholder_influence_mapping_tool_english.pdf. (Accessed 25 May 2021).
[77] I. Hovland, Successful communication a toolkit for researchers and civil society organisations [Online]. Available, Overseas Development Institute, 2005, https://cdn.odi.org/media/documents/192.pdf. (Accessed 26 May 2021).
[78] R.M. Shrestha, J.S. Acharya, Sustainable Energy Access Planning: A Framework, Asian Development Bank, Mandaluyong City, Metro Manila, Philippines, 2015.
[79] S.C. Funnell, P.J. Rogers, Purposeful Program Theory: Effective Use of Theories of Change And Logic Models vol. 31, John Wiley & Sons, San Francisco, 2011.
[80] W.K. Kellogg Foundation, Logic Model Development Guide: Using Logic Models to Bring Together Planning, Evaluation, And Action, W.K. Kellogg Foundation, Battle Creek, Michigan, 2004.
[81] EssentialTech Centre, The impact canvas [Online]. Available, EPFL CODEV/EssentialTech Centre, 2017, https://essentialtech.center/wp-content/uploads/2019/03/ImpactCanvas_PDF_Tool.pdf.
[82] P. Sandwell, N. Ekins-Daukes, J. Nelson, What are the greatest opportunities for PV to contribute to rural development? Energy Procedia 130 (2017) 139–146, https://doi.org/10.1016/j.egypro.2017.09.416.
[83] P. Sandwell, S. Wheeler, J. Nelson, Supporting Rural Electrification in Developing Countries - CLOVER: Modelling Minigrids for Rural Electrification, Project Note, Grantham Institute, Imperial College London, London, 2017 [Online]. Available: https://www.imperial.ac.uk/media/imperial-college/grantham-institute/public/publications/briefing-papers/Supporting-Rural-Electrification-in-Developing-Countries—CLOVER-Modelling-Minigrids.pdf.
[84] P. Sandwell, Assessing the Potential for Photovoltaic Technology to Mitigate Greenhouse Gas Emissions: Life Cycle Analysis And Energy Systems Modelling for Electricity Access in Rural India, Imperial College London, London, 2017, https://doi.org/10.25560/58881.
[85] P. Sandwell, CLOVER, GitHub, London, 2018.
[86] A. Osterwalder, Y. Pigneur, M.A.-Y. Oliveira, J.J.P. Ferreira, Business Model Generation: a handbook for visionaries, game changers and challengers, Afr. J. Bus. Manag. 5 (7) (2011) 22–30.
[87] S. Makohliso, et al., Medical Technology Innovation for a Sustainable Impact in Low- And Middle-Income Countries: A Holistic Approach, Jul. 2020, https://doi.org/10.31224/osf.io/2dytg. Toolkit.
[88] A. Joyce, R.L. Paquin, The triple layered business model canvas: a tool to design more sustainable business models, J. Clean. Prod. 135 (Nov. 2016) 1474–1486, https://doi.org/10.1016/j.jclepro.2016.06.067.
[89] L. Bloom, D.A. Betts, The two worlds of humanitarian innovation, University of Oxford, Oxford, 2013. https://www.rsc.ox.ac.uk/files/files-1/wp94-two-worlds-humanitarian-innovation-2013.pdf.
[90] A. Betts, L. Bloom, N. Weaver, Refugee Innovation: humanitarian innovation that starts with communities, University of Oxford, Oxford, Jul. 2015. https://www.rsc.ox.ac.uk/refugee-innovation-humanitarian-innovation-that-starts-with-communities.
[91] S. Rosenberg-Jansen, M. Barlow, S. Peisch, N. Ponnan, P. Rathi, 'Sustainable Humanitarian Energy Services Inclusive participation, lessons learnt, and paths forward'. Practical Action [Online]. Available, https://infohub.practicalaction.org/bitstream/handle/11283/620670/Policybrief_7_web_updated.pdf?sequence=1&isAllowed=y, 2018.
[92] B. Garside, S. Wykes, Planning Pro-poor Energy Services for Maximum Impact: The Energy Delivery Model Toolkit, CAFOD & IIED, London, 2017.
[93] Mercy Corps, Inclusive energy access handbook [Online]. Available, Mercy Corps, 2020, https://www.womensrefugeecommission.org/wp-content/uploads/2020/10/Inclusive-Energy-Access-Handbook-2020.pdf. (Accessed 26 May 2021).
[94] SELCO Foundation, Problem Assessment Toolkit, SELCO Foundation, Jan. 2013.
[95] Norwegian Refugee Council, International Organization for Migration, NRC community coordination toolbox [Online]. Available, Norwegian Refugee Council, 2020, https://cct.nrc.no/chapter/1. (Accessed 2 November 2021).
[96] J. Noble, Theory of change in ten steps [Online]. Available, New Philanthropy Capital (NPC), London, 2019, https://www.thinknpc.org/wp-content/uploads/2019/10/Theory-of-Change-10-Steps.pdf. (Accessed 20 August 2020).
[97] T. Lambert, P. Gilman, P. Lilienthal, Micropower system modeling with HOMER, in: Integration of Alternative Sources of Energy 1, 2006, pp. 379–385, no. 1.
[98] United States National Renewable Energy Laboratory (NREL), HOMER Powering Health Tool [Online]. Available, https://poweringhealth.homerenergy.com/, 2021.
[99] A. Osterwalder, Y. Pigneur, C.L. Tucci, Clarifying business models: origins, present, and future of the concept, CAIS 16 (2005), https://doi.org/10.17705/1CAIS.01601.
[100] EssentialTech Centre, The social business model canvas, EPFL CODEV/EssentialTech Centre, 2017. https://essentialtech.center/wp-content/uploads/2019/03/Social_BM_Canvas_PDF_Tool.pdf.
[101] A. Maurya, Running Lean: Iterate From Plan A to a Plan That Works, O'Reilly Media, Inc., 2012.
[102] T. Ittelson, Financial Statements: A Step-by-step Guide to Understanding And Creating Financial Reports, 2009.
[103] D. Ross, F. Simonsen, How Do We Finance It? A Guide to Financing of Mini-grid Companies And Projects, Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH, Bonn and Eschborn, Germany, 2017.
[104] World Bank, Data: Djibouti, World Bank, 2021. https://data.worldbank.org/country/djibouti. (Accessed 21 May 2021).
[105] UNHCR, Djibouti, UNHCR Global Focus, 2021. Djibouti Report, https://reporting.unhcr.org/djibouti. (Accessed 29 July 2020).
[106] UNHCR, Comprehensive Refugee Response Framework: delivering more comprehensive and predictable responses for refugees, UNHCR, 2021. https://www.unhcr.org/comprehensive-refugee-response-framework-crrf.html. (Accessed 25 May 2021).
[107] UNITAR, Sustainable Energy for Humanitarian Settings in Djibouti: Rapid Energy Assessment, United Nations Institute for Training And Research (UNITAR) and Global Plan of Action (GPA) for Sustainable Energy Solutions in Situations of Displacement, Geneva, Feb. 2020.
[108] UNITAR, Project Proposal: Delivering Sustainable Energy Solutions to Djibouti Operations And Improving Income Generating Activities for the Displaced, United Nations Institute for Training and Research, 2019.
[109] Republic of Djibouti, Vision Djibouti 2035, Republic of Djibouti, 2014. http://www.ccd.dj/w2017/wp-content/uploads/2016/01/Vision-Nationale.pdf.
[110] IRENA, Energy profile: Djibouti [Online]. Available, IRENA, Nov. 29, 2021, https://www.irena.org/IRENADocuments/Statistical_Profiles/Africa/Djibouti_Africa_RE_SP.pdf.
[111] USAID, Djibouti Power Africa Fact Sheet, USAID, 2020. https://www.usaid.gov/powerafrica/djibouti. (Accessed 23 November 2020).
[112] UNHCR, Livelihoods Strategic Plan UNHCR Djibouti 2018-2022, UN High Commissioner for Refugees, 2018.
[113] SELCO Foundation, Sustainable Energy And Livelihoods Ecosystems for Refugee Communities in Djibouti, SELCO Foundation, 2020. Djibouti Report.
[114] SELCO Foundation, Sustainable energy ecosystems for livelihood generation and integration of refugee communities in Djibouti [Online]. Available, Global SDG7 Hubs, 2020, https://globalsdg7hubs.org/wp-content/uploads/2020/12/PPT-post-Djibouti.pdf. (Accessed 24 December 2020).
[115] E.C.X. Ikejemba, The empirical failures of attaining the societal benefits of renewable energy development projects in sub-Saharan Africa, Renew. Energy (2020) 9.
[116] UNICEF, Djibouti humanitarian situation report - mid-year 2018 [Online]. Available, United Nations Children's Fund, 2018, https://www.unicef.org/appeals/files/UNICEF_Djibouti_Humanitarian_Situation_Report_MidYear_2018.pdf. (Accessed 21 August 2020).
[117] UNHCR, ILO, Refugee Local Economic Integration - Djibouti: Market Assessment for Refugee Local Economic Integration in Djibouti, United Nations High Commissioner for Refugees and International Labour Organization, Djibouti, 2017.
[[118] IMF, Djibouti: staff report for the 2019 article iv consultation— debt sustainability analysis, International Monetary Fund (IMF), Washington DC, 2019.
[119] SELCO Foundation, Sustainable energy and livelihood: a collection of 65 livelihood applications [Online]. Available, SELCO Foundation, 2019, https://selcofoundation.org/wp-content/uploads/2021/05/SELCO-Foundation_-65-livelihoods-solutions-from-ground.pdf.
[120] IRENA, Djibouti Renewable Readiness Assessment, International Renewable Energy Agency, Abu Dhabi, 2015.
[121] GPA Coordination Unit, Blended finance toolkit for clean energy solutions in humanitarian and displacement settings, UNITAR, Geneva, Switzerland, Jan. 2022. https://www.humanitarianenergy.org/news/latest/blended-finance-solutions-bridging-the-funding-gap-and-driving-self-sustaining-solutions-in-displacement-settings.
[122] D. Fouquet, Y. Traum, R. Skehan, Identification and analysis of standard clauses of PPA and leasing agreements for energy provision in the humanitarian sector [Online]. Available, Becker Büttner Held, Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH (GIZ) and the Global Plan of Action for Sustainable Energy Solutions in Situations of Displacement (GPA), Brussels, 2020, https://energypedia.info/images/2/21/Identification_And_Analysis_Of_Standard_Clauses_Of_PPA_And_Leasing_Agreements_For_Energy_Provision_In_The_Humanitarian_Sector.pdf.
[123] J.C. Hartman, I.C. Schafrick, The relevant internal rate of return, Eng. Econ. 49 (2) (2004) 139–158.
[124] Y. Zhang, C. Ochieng, Understanding cookstove and fuel stacking to achieve SDG7, World Bank Blogs, Nov. 22, 2020. https://blogs.worldbank.org/energy/understanding-cookstove-and-fuel-stacking-achieve-sdg7. (Accessed 19 November 2021).
[125] G. Hoffstaedter K. Koizumi, Urban refugees: challenges in protection, services and policy. Abingdon: Routledge.
[126] V.C. Broto, et al., A research agenda for a people-centred approach to energy access in the urbanizing global south, Nat. Energy 2 (10) (Oct. 2017) 776–779, https://doi.org/10.1038/s41560-017-0007-x.
[127] NORCAP, Urban-A, Urban transitions: clean energy in urban recovery [Online]. Available, Norwegian Refugee Council, Oslo, Norway, Dec. 2021, https://www.nrc.no/resources/reports/urban-transitions-clean-energy-in-urban-recovery/. (Accessed 28 February 2022).

[128] J. Haselip, K. Chen, H. Marwah, E. Puzzolo, Cooking in the margins: exploring the role of liquefied petroleum gas for refugees in low-income countries, Energy Res. Soc. Sci. 83 (Jan. 2022), 102346, https://doi.org/10.1016/j.erss.2021.102346.
[129] C.I. Med, Multidisciplinarity, interdisciplinarity and transdisciplinarity in health research, services, education and policy: 1. Definitions, objectives, and evidence of effectiveness, Clin. Invest. Med. 29 (6) (2006) 351–364.
[130] M. Montedonico, F. Herrera-Neira, A. Marconi, A. Urquiza, R. Palma-Behnke, Co-construction of energy solutions: lessons learned from experiences in Chile, Energy Res. Soc. Sci. 45 (Nov. 2018) 173–183, https://doi.org/10.1016/j.erss.2018.08.004.
[131] S. Thomas, M. Richter, W. Lestari, S. Prabawaningtyas, Y. Anggoro, I. Kuntoadji, Transdisciplinary research methods in community energy development and governance in Indonesia: insights for sustainability science, Energy Res. Soc. Sci. 45 (Nov. 2018) 184–194, https://doi.org/10.1016/j.erss.2018.06.021.
[132] Global Platform for Action on Sustainable Energy in Displacement Settings, Humanitarian energy exchange network, Global Platform for Action on Sustainable Energy in Displacement Settings, Jan. 25, 2022. https://www.humanitarianenergy.org/thematic-working-areas/humanitarian-energy-exchange-network/. (Accessed 23 February 2022).
[133] J. Terrapon-Pfaff, M.-C. Gröne, C. Dienst, W. Ortiz, Impact pathways of small-scale energy projects in the global south – findings from a systematic evaluation, Renew. Sustain. Energy Rev. 95 (Nov. 2018) 84–94, https://doi.org/10.1016/j.rser.2018.06.045.
[134] S. Samarakoon, A justice and wellbeing centered framework for analysing energy poverty in the Global South, Ecol. Econ. 165 (Nov. 2019), 106385, https://doi.org/10.1016/j.ecolecon.2019.106385.
[135] B.K. Sovacool, M.H. Dworkin, Energy justice: conceptual insights and practical applications, Appl. Energy 142 (Mar. 2015) 435–444, https://doi.org/10.1016/j.apenergy.2015.01.002.
[136] UNITAR, WFP, Introduction to energy access in displacement contexts [Online]. Available, United Nations Institute for Training and Research and World Food Programme, 2021, https://360.articulate.com/review/content/06b14127-66e4-46e0-af05-409f354fa6ad/review. (Accessed 30 May 2021).